The sound of dynamical dark energy and modified gravity

João Rebouças,[1,2] Guilherme Brando,[2] Felipe T. Falciano,[2] and Vivian Miranda[3]

[1] *Department of Astronomy/Steward Observatory, University of Arizona, 933 North Cherry Avenue, Tucson, AZ 85721, USA*

[2] *CBPF - Brazilian Center for Research in Physics, Xavier Sigaud st. 150, zip 22290-180, Rio de Janeiro, RJ, Brazil**

[3] *C. N. Yang Institute for Theoretical Physics, Stony Brook University, Stony Brook, NY, 11794, USA*

(Dated: October 21, 2026)

Different candidate models are able to reproduce the dynamical dark energy signal preferred by combinations of recent distance measurements. These models may be distinguished by the behavior of their perturbations, which are controlled by the effective sound speed $c_s^2(k,a)$. To explore correlations between the dark energy sound speed and [**Edit**: dynamical] behavior, we test modified gravity (MG) scenarios in which the dark energy equation of state and sound speed determine modifications to the clustering of matter. We investigate the impact of varying the dark energy sound speed on several cosmological quantities in both General Relativity (GR) and MG. We constrain the dark energy and modified gravity parameters using measurements of the Cosmic Microwave Background (CMB) from Planck PR4, type Ia supernova luminosity distances (SN) from [**Edit**: DES-Dovekie], Baryon Acoustic Oscillations (BAO) from DESI DR2, and cosmic shear from DES-Y3. Using the combination of CMB+BAO+SN, we find that, in the MG scenarios, the preference for dynamical dark energy is correlated with deviations from GR over redshifts $z < 2$ at over 95% confidence level. The significance of these deviations persists when considering a dynamical or superluminal sound speed, but vanishes if we assume a cosmological constant. Including cosmic shear and CMB lensing data significantly shifts the constraints towards GR. Our framework enables the exploration of modified gravity models using the dark energy sound speed as a physically meaningful free parameter.

I. INTRODUCTION

One of the main scientific goals of modern cosmology is to uncover the nature of dark energy. In the ΛCDM model, the current paradigm, dark energy is assumed to be a cosmological constant driving the present accelerated expansion of the Universe. Despite its simplicity, ΛCDM provides a remarkably good fit to a wide range of independent measurements, including Cosmic Microwave Background (CMB) anisotropies [1–6], correlation functions of galaxy positions and shapes [7–15], and the luminosity distance-redshift relation of type Ia supernovae [16–22]. In recent years, with the rise of precision cosmology, ΛCDM has faced several challenges, most notably the Hubble tension between CMB and cosmic distance ladder measurements of the H_0 parameter [23, 24]. A milder discrepancy has also been reported by galaxy and CMB surveys involving the S_8 parameter, which describes the amplitude of matter fluctuations, although its significance depends on the specific datasets and systematic uncertainty modelling [7, 25–28].

More recently, new distance measurements from type Ia supernovae [17, 19–21] and galaxy surveys using the Baryon Acoustic Oscillation (BAO) feature [8–10] have revealed another potential inconsistency in the ΛCDM model, this time favoring a dynamical dark energy fluid over a cosmological constant. These results are based on the w_0w_a model [29, 30], in which the dark energy equation of state is a linear function of the scale factor, $w(a) = w_0 + w_a(1-a)$. This scenario has been widely used as a null test of ΛCDM, since the model reduces to a cosmological constant when $w_0 = -1$ and $w_a = 0$. While past analyses were able to recover the cosmological constant within the error bars [1, 16], combinations of state-of-the-art datasets from the DESI, Union3, DES, Pantheon+, and Planck collaborations exclude the cosmological constant with statistical significance ranging from 2.8σ to 4.2σ [9, 20]. The preferred dark energy behavior has a phantom ($w_{\rm DE} < -1$) phase in the past, smoothly transitioning to a non-phantom ($w_{\rm DE} > -1$) phase at redshift $z \approx 0.5$ [31–33]. Whether these results represent the first hint of dynamical dark energy remains under debate [34–36].

Connecting the phenomenological w_0w_a model to a fundamental theory is challenging, and multiple candidates have been proposed to explain the phantom crossing behavior [37–46]. The simplest fundamental models assume that dark energy is an additional, minimally coupled scalar field, as in quintessence [47, 48] and k-essence [49] theories. The former cannot have a phantom equation of state, while the latter is plagued by instabilities in their perturbations when the equation of state is phantom [50–53]. More general scalar-tensor theories, such as the Effective Field Theory of Dark Energy (EFTofDE) [54–56] and Horndeski theory [57], naturally introduce new couplings between the metric tensor and the scalar field in the action, and the phenomenology associated with these extra couplings can be characterized by free functions of time [58]. At the same time, this extra freedom significantly enlarges the theory space, mak-

* joaoreboucas@arizona.edu

ing it difficult to explore viable dark energy candidates.

While investigating modified gravity theories as possible candidates for the current acceleration of our Universe is a more first-principles-oriented approach, model-agnostic investigations remain highly valuable, especially in the era of precision cosmology, since they allow us to perform more targeted stress tests of the concordance model and to search for possible cracks within it. Within this approach, it is useful to adopt a fluid description of dark energy while going beyond the smooth dynamical dark energy paradigm, in which the dark energy fluid is described only by its equation of state. Instead, one may also allow the sound speed of the fluid to vary. It is well known that, at the perturbative level, a general perfect fluid can be characterized by its sound speed, c_s^2, which may depend on both time and scale, as well as by its viscosity parameter, $c_v^2(k,a)$ [59, 60]. Quintessence models are characterized by $c_s^2 = 1$, which suppresses perturbations within the cosmological horizon, and $c_v^2 = 0$, implying dark energy has no anisotropic stress. These are standard assumptions adopted in most analyses of fluid dark energy models.

In k-essence and EFTofDE scenarios, however, the sound speed may deviate significantly from $c_s^2 = 1$, allowing dark energy perturbations to contribute nontrivially to the gravitational potential. Consequently, the impact of dark energy perturbations on large-scale structure (LSS) can be used to distinguish between candidate models that otherwise predict the same background expansion history of the Universe [61–65]. The difficulty in constraining the dark energy sound speed stems from the fact that its effects become relevant only on very large scales compared to models with $c_s^2 = 1$. Previous studies have shown that the impact of the late-time dark energy sound speed on cosmological observables is generally small [66–70]. This conclusion, however, may not hold in modified gravity scenarios, where the effective dark energy sound speed can be sourced by additional terms in the gravitational action that modify the clustering properties of matter [58, 71, 72].

In this work, we propose a new parametrized approach to testing the dark energy sound speed by investigating its impact on the growth of cosmic structure. As mentioned above, simple minimally coupled scalar-field models such as quintessence and k-essence are generically prone to instabilities when crossing the phantom divide, $w_{\rm DE} = -1$. In Ref. [73], it was shown that introducing self-interaction terms in the scalar-field action, commonly referred to as kinetic gravity braiding (KGB), causes the stress-energy tensor to deviate from the perfect-fluid form. This feature allows the scalar field to cross the phantom divide without developing instabilities.

In this class of models, deviations from General Relativity are characterized by two time-dependent functions, $\alpha_{\rm K}$ and $\alpha_{\rm B}$, which are directly related to the scalar-field sound speed. Our goal is therefore to move away from the standard covariant approach to modified gravity, in which one specifies functional forms for $\alpha_{\rm K}$ and $\alpha_{\rm B}$, and instead adopt a phenomenological perspective in which the dark energy sound speed is treated as the fundamental varying quantity. In practice, we still parameterize the kineticity function, $\alpha_{\rm K}$, while deriving $\alpha_{\rm B}$ from the imposed sound-speed evolution. A similar approach was recently employed in [74], where a general time evolution for c_s^2 was assumed. This approach ensures that dark energy perturbations remain stable even with a phantom equation of state. In this modified gravity scenario, we use data from CMB, BAO, supernovae, and cosmic shear, to obtain novel constraints on the dark energy equation of state, sound speed, and deviations from General Relativity (GR).

This paper is organized as follows: Section II discusses the theoretical background of the sound speed in scalar-field theories and presents our phenomenological relation; Section III describes the datasets used in our work and the methodology used to extract cosmological parameter constraints from data; Section IV presents and discusses the results of the data analysis; and Section V summarizes the discussion and our findings.

II. THEORY

A. Fluid Dark Energy

We assume a perturbed FRW Universe in the scalar Newtonian gauge, neglecting vector and tensor perturbations,

$$ds^2 = a^2(\tau)\left[-(1+2\Psi)d\tau^2 + (1-2\Phi)\delta_{ij}dx^i dx^j\right]. \quad (1)$$

We parameterize the dark energy energy-momentum tensor as

$$T^0{}_0 = -\rho - \delta\rho, \quad (2a)$$
$$T^i{}_0 = -(\rho + P)v^i, \quad (2b)$$
$$T^i{}_j = \delta^i{}_j(P + \delta P) + P\Pi^i{}_j, \quad (2c)$$

where ρ and $P = w_{\rm DE}\rho$ are the background density and pressure, $\delta\rho$ is the density perturbation, δP is the pressure perturbation, v^i is the bulk velocity, and Π^i_j is the anisotropic stress. The energy-momentum conservation equations do not determine the evolution of δP, so an additional parameterization is needed to fully specify the microphysics of the dark energy fluid. At the linear perturbation level, one such parameterization is defined as follows. The pressure perturbation can be decomposed into adiabatic and non-adiabatic (or entropic) parts,

$$\delta P = \frac{P'}{\rho'}\delta\rho + \delta P_s, \quad (3)$$

[**Edit**: where primes denote derivatives with respect to conformal time τ]. The first term on the right-hand side is the adiabatic pressure perturbation, and the second

is the entropic pressure perturbation, which is gauge-independent. Therefore, δP_s must be parameterized in terms of gauge-independent perturbations. One simple but widely adopted model is [59]

$$\delta P_s = c_s^2 \left[\delta\rho + 3\mathcal{H}(\rho + P)\frac{v}{k}\right], \tag{4}$$

where c_s^2 is a free function of the scale factor a and Fourier wavenumber k, and the term inside square brackets is the comoving density perturbation, which is gauge-independent. This effective sound speed parameterization is recovered in scalar field theories.

The standard energy-momentum conservation equations are unstable if the equation of state is dynamical and happens to cross $w = -1$ at some redshift. An alternative for the fluid perturbation equations is the Parameterized Post-Friedmann (PPF) approach [75], a reparameterization of the perturbation variables that has been shown to agree with the fluid equations with great precision and can be applied to phantom crossing fluids. The new variable Γ is a proxy to the dark energy density perturbations in the matter rest frame, and its evolution equation is given by

$$(1 + c_\Gamma^2 k_H^2)[\Gamma' + (1 + c_\Gamma^2 k_H^2)\Gamma] = \frac{4\pi G(1 + w_{\rm DE})\rho_{\rm DE} v_T}{k_H H^2}, \tag{5}$$

where $k_H = k/(aH)$, v_T is the total bulk velocity of all species except for dark energy, and c_Γ is a parameter controlling the transition scale where dark energy becomes smooth with respect to dark matter. Ref. [75] sets $c_\Gamma = 0.4 c_s$ to match predictions from phantom crossing scalar field models.

B. Modified Gravity

Horndeski theory provides the most general scalar-tensor gravity model with second-order equations of motion [57], encompassing many models for dark energy from quintessence [47, 48, 76] and k-essence [49] to more substantial modifications of gravity such as galileons [77, 78] or $f(R)$ gravity [79, 80]. The theory is summarized by the general Lagrangian,

$$S = \int d^4x\sqrt{-g}\left(\sum_{i=2}^{5}\mathcal{L}_i + \mathcal{L}_m\right), \tag{6}$$

where $\mathcal{L}_m$ is the matter Lagrangian and

$$\mathcal{L}_2 = G_2, \tag{7a}$$
$$\mathcal{L}_3 = -G_3\Box\phi, \tag{7b}$$
$$\mathcal{L}_4 = G_4 R + G_{4,X}[(\Box\phi)^2 - \nabla_\mu\nabla_\nu\phi\nabla^\mu\nabla^\nu\phi], \tag{7c}$$
$$\mathcal{L}_5 = G_5 G_{\mu\nu}\nabla^\mu\nabla^\nu\phi - \frac{1}{6}G_{5,X}[(\Box\phi)^3 + 2\nabla^\mu\nabla_\nu\phi\nabla^\nu\nabla_\rho\phi\nabla^\rho\nabla_\mu\phi - 3\nabla_\mu\nabla_\nu\phi\nabla^\mu\nabla^\nu\phi\Box\phi], \tag{7d}$$

where $G_i(\phi, X)$ are arbitrary functions of the scalar field ϕ and its kinetic term $X = -\partial_\mu\phi\partial^\mu\phi/2$, and commas represent partial derivatives. We adopt the parameterization from Ref. [58], where the linear cosmological perturbations can be determined by the background expansion history and four additional functions of time: $\alpha_{\rm K}$, $\alpha_{\rm B}$, $\alpha_{\rm M}$ and $\alpha_{\rm T}$; their specific definitions can be found in Appendix A of [58]. In this case, the dark energy sound speed is given by [58, 64, 71, 72, 74]

$$c_s^2 = \frac{1}{D_{\rm kin}}\left[(2 - \alpha_{\rm B})\left(-\frac{H'}{aH^2} + \frac{\alpha_{\rm B}}{2}(1 + \alpha_{\rm T}) + \alpha_{\rm M} - \alpha_{\rm T}\right) - \frac{8\pi G(\rho_{\rm no\,DE} + P_{\rm no\,DE})}{H^2 M_*^2} + \frac{\alpha_{\rm B}'}{aH}\right], \tag{8}$$

where $D_{\rm kin} = \alpha_{\rm K} + 3\alpha_{\rm B}^2/2$, $\alpha_{\rm M} = d\ln M_*^2/d\ln a$ and the subscript "no DE" refers to the sum of all species excluding dark energy.

The function $\alpha_{\rm T}$ controls the deviation of the sound speed of gravitational waves in these theories, and it has been tightly constrained [81]. Within our work, we will assume $\alpha_{\rm T} = 0$. For simplicity, we further assume $\alpha_{\rm M} = 0$: this assumption is compatible with a minimally coupled scalar field with $G_4 = 1/2$ and $G_5 = 0$. With these assumptions, the slip parameter $\eta = \Phi/\Psi$ is equal to unity, indicating the absence of dark energy anisotropic stress. While anisotropic stress is an important component of the dark energy microphysics, we leave the case of $\alpha_{\rm M} \neq 0$ for future work.

C. Sound Speed and Growth

While standard analyses of modified gravity parameterize the functions α_i [82], we take a different approach: we invert Equation 8 to obtain a differential equation for the braiding function $\alpha_{\rm B}$ in terms of $\alpha_{\rm K}$ and c_s^2. A similar approach was employed in Ref. [74], where both c_s^2 and $D_{\rm kin}$ were parameterized using a Gaussian Process model on top of power-law parametrizations. After setting $\alpha_{\rm M} = \alpha_{\rm T} = 0$, we obtain

$$\frac{d\alpha_{\rm B}}{d\ln a} = c_s^2\left(\alpha_{\rm K} + \frac{3}{2}\alpha_{\rm B}^2\right) + \frac{\alpha_{\rm B}^2}{2} - \alpha_{\rm B}\left(\frac{d\ln H}{d\ln a} + 1\right) - 3(1 + w_{\rm DE})\Omega_{\rm DE}(a), \tag{9}$$

where $\Omega_{\rm DE}(a) = 8\pi G\rho_{\rm DE}(a)/3H^2$. Given an initial condition $\alpha_{{\rm B},i}$ at a high redshift z_i and specific functional forms for c_s^2 and $\alpha_{\rm K}$, one can fully determine the function $\alpha_{\rm B}$. Throughout this work, we choose an initial redshift $z_i = 10^5$ with initial condition $\alpha_{\rm B}(z_i) = 0$.

We consider four parametrizations for $\alpha_{\rm K}$:

- Proportional to $\alpha_{\rm B}$:

$$\alpha_{\rm K} = 3\lambda_{\rm MG}\,\alpha_{\rm B}; \tag{10}$$

- Proportional to the dark energy abundance:

$$\alpha_{\rm K} = \lambda_{\rm MG}\frac{\Omega_{\rm DE}(a)}{\Omega_{\rm DE,0}}; \tag{11}$$

- K-essence-like:

$$\alpha_{\rm K} = \frac{\Omega_{\rm DE}(a)(1+w_{\rm DE})}{c_s^2}; \tag{12}$$

inspired by k-essence models [58];

- Cubic Galileon-like:

$$\alpha_{\rm K} = 6\lambda_{\rm MG}\frac{H_0^4\,\Omega_{\rm DE}(a=1)}{H(a)^4}, \tag{13}$$

motivated by Cubic Galileon theories [77, 78, 83].

The first two parametrizations are purely phenomenological, while the latter two are motivated by scalar–tensor theories, since they reproduce the functional form of $\alpha_{\rm K}$ appearing in their covariant formulations. We refer to them as "k-essence-like" and "Cubic Galileon-like" because only the form of $\alpha_{\rm K}$ is inherited from these models, while the functions $\alpha_{\rm B}$ and $w_{\rm DE}$ differ fundamentally from those of their covariant counterparts. In particular, k-essence models satisfy $\alpha_{\rm B} = 0$, while in Cubic Galileon theories one has $\alpha_{\rm B} = \alpha_{\rm K}/3$; the latter motivates us to parameterize the first case as $\alpha_{\rm K} = 3\lambda_{\rm MG}\,\alpha_{\rm B}$. Furthermore, in the Cubic Galileon-like parametrization we also introduce the amplitude parameter $\lambda_{\rm MG}$ in order to test possible deviations from the theoretical prediction, such that $\lambda_{\rm MG} = 1$ recovers the standard Cubic Galileon form for $\alpha_{\rm K}$. We do not introduce an additional rescaling parameter in the k-essence-like case, since variations in c_s^2 already control the overall amplitude of $\alpha_{\rm K}$.

For the dark energy sound speed, we consider two cases: a constant sound speed and a time-dependent parametrization,

$$c_s^2(a) = c_{s,0}^2 + c_{s,a}^2(1-a), \tag{14}$$

analogous to the commonly adopted CPL parametrization for the dark energy equation of state,

$$w_{\rm DE}(a) = w_0 + w_a(1-a). \tag{15}$$

Within this modified gravity framework, we may also allow for superluminal dark energy sound speeds, *i.e.* $c_s^2 > 1$. Although in the present framework the dark energy stress-energy tensor takes the form of an imperfect fluid [73], we still expect this parametrization to capture the overall time evolution of c_s^2 in a sufficiently general manner.

Given our assumptions of $\alpha_{\rm T} = \alpha_{\rm M} = 0$, the braiding and kineticity functions alter the Einstein equations for the metric as [71]

$$k^2\Psi = -4\pi G_N\mu([\textbf{Edit}: k,]a)a^2\rho\delta, \tag{16a}$$
$$k^2(\Phi+\Psi) = -8\pi G_N\Sigma([\textbf{Edit}: k,]a)a^2\rho\delta, \tag{16b}$$

where G_N is Newton's gravitational constant, [**Edit**: and μ and Σ are scale-dependent functions that absorb the modified gravity effects in the Einstein equations. Given our assumptions of $\alpha_{\rm M} = \alpha_{\rm T} = 0$, $\mu = \Sigma$, *i.e.* there is no gravitational slip. The μ function is given by [58, 83]]

$$\mu(k,a) = \frac{\mu_p(a) + k_{\mathcal{H}}^2\left(c_{s,\rm N}^2 + \frac{\alpha_B^2}{2}\right)}{\mu_p(a) + k_{\mathcal{H}}^2 c_{s,\rm N}^2}, \tag{17}$$

[**Edit**: where $k_{\mathcal{H}} = k/(aH)$, $c_{s,N}^2$ is the numerator from Equation 8 and]

$$\mu_p = \frac{3}{4H^2M_{\rm pl}^2}\left\{\left[2c_{sN}^2 + \alpha_B(\alpha_B-2)\right](\rho+p) + 2\alpha_B\frac{dp}{d\ln a}\right\}. \tag{18}$$

[**Edit**: Notice that, in the deep subhorizon limit, $k_{\mathcal{H}} \to \infty$, this equation tends to the quasi-static approximation expression,]

$$\mu(a) = 1 + \frac{\alpha_{\rm B}^2}{2c_{s,\rm N}^2}. \tag{19}$$

Perturbations are stable if and only if $c_s^2 > 0$ and $D_{\rm kin} = \alpha_{\rm K} + 3\alpha_{\rm B}^2/2 > 0$ [58, 71]. The first condition can be straightforwardly enforced by imposing the prior $c_s^2 > 0$. The second condition is trivially satisfied if $\alpha_{\rm K} > 0$, which holds if we assume $\alpha_{\rm K} \propto \Omega_{\rm DE}$ or the Cubic Galileon-like parametrization with $\lambda_{\rm MG} > 0$. For the k-essence-like parametrization, perturbations are stable if $w > -1$. Stability can still be achieved in the phantom regime if $3\alpha_{\rm B}^2/2 > |\alpha_{\rm K}|$, but this condition is automatically violated for the initial condition $\alpha_{\rm B} = 0$. Therefore, when analyzing k-essence-like models, we impose $w > -1$. In the case $\alpha_{\rm K} \propto \alpha_{\rm B}$, it is not trivial to ensure stability a priori. Therefore, during the numerical integration of Equation 9, we check the value of $D_{\rm kin}$ and raise an error if it violates the stability condition, effectively excluding unstable models from the analysis.

We implemented Equations 9 and 16 in a modified version of CAMB[1]. After solving Equation 9, we compute μ using Equation **??**, with the replacement

$$c_{s,\rm N}^2 = c_s^2\left(\alpha_{\rm K} + \frac{3}{2}\alpha_{\rm B}^2\right).$$

An important consequence of Equation **??** is that the GR value, $\mu(a) = 1$, can only be recovered if either $\alpha_{\rm B} = 0$, corresponding to a pure k-essence theory, or if $c_{s,\rm N}^2 \to \infty$. In this work, however, we treat c_s^2 as a free parameter, while $D_{\rm kin}$ is fixed by the evolution of $\alpha_{\rm K}$ and $\alpha_{\rm B}$. Therefore, in all physical scenarios considered here, the combination $c_s^2 D_{\rm kin}$ remains finite. As a result, the recovery of GR-like growth should be understood as the limit in which deviations from $\mu = 1$ become small, rather than as an exact return to GR.

Figure 1 shows solutions for μ for several values of $\lambda_{\rm MG}$ and c_s^2, assuming $w_0 = -0.838$ and $w_a = -0.62$. For the

[1] https://github.com/joaoreboucas1/CAMB-cs2

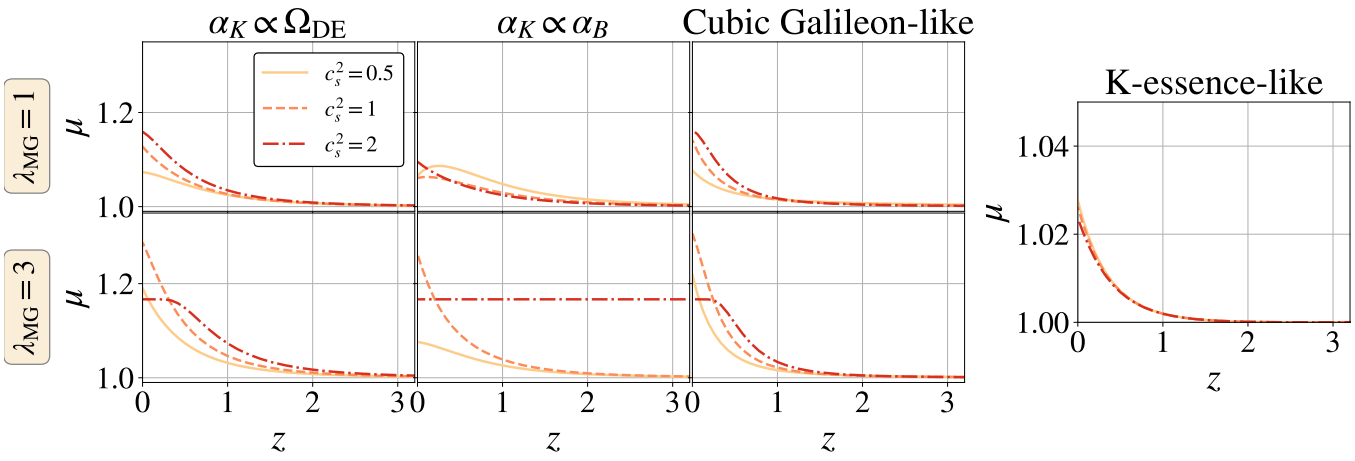


FIG. 1. Solutions for μ from Equation 9 for several values of c_s^2 and $\lambda_{\rm MG}$, for the four parameterizations of $\alpha_{\rm K}$ considered, assuming $w_0 = -0.838$ and $w_a = -0.62$. The k-essence-like case is not parameterized by $\lambda_{\rm MG}$, and we assume $w_a = -0.162$ to stabilize the perturbations. [**Edit**: In the case of $\alpha_{\rm K} \propto \alpha_{\rm B}$] with $c_s^2 = 2$ and $\lambda_{\rm MG} = 3$, μ starts deviating from unity around $z = 20$.

k-essence-like case, we assume $w_a = -0.162$ in order to stabilize the perturbations. In typical cases, μ is close to unity at redshifts $z > 5$, similar to other phenomenological models for modified gravity in which $\mu \propto \Omega_{\rm DE}$ [84–87]. Furthermore, since our priors strongly favor $\alpha_{\rm K} > 0$, it follows from Equation **??** that $\mu \geq 1$. This qualitative behavior is shared by the four parameterizations of $\alpha_{\rm K}$. The strength of the deviation from $\mu = 1$ and the redshift at which this deviation becomes significant depend on the values of c_s^2 and $\lambda_{\rm MG}$. In general, higher values of $\lambda_{\rm MG}$ induce stronger and earlier deviations from GR. Figure 1 also shows that increasing c_s^2 leads to earlier deviations from GR. On the other hand, Equation **??** implies that, at fixed α_B and $D_{\rm kin}$, μ is inversely proportional to c_s^2.

From Equation 9, the late-time evolution of $\alpha_{\rm B}$ is tied to the dark energy equation of state. Interestingly, for the $\alpha_{\rm K} \propto \alpha_{\rm B}$ and k-essence-like parameterizations, $\alpha_{\rm B}$ is equal to zero if dark energy is a cosmological constant, leading to $\mu = 1$ and no deviations from GR. For the other parametrizations, the deviations from GR are reduced when we consider a cosmological constant rather than dynamical dark energy. This is why the k-essence-like case in Figure 1 shows weaker deviations from GR.

In more extreme cases, such as the one with $\alpha_{\rm K} \propto \alpha_{\rm B}$, $c_s^2 = 2$ and $\lambda_{\rm MG} = 3$ shown in Figure 1, μ may deviate from unity at higher redshifts. In that specific case, this happens at $z \approx 20$. For higher values of c_s^2 and $\lambda_{\rm MG}$, the evolution of μ may happen even earlier, potentially affecting recombination. In the following analysis, we confirmed that the models preferred by data do not alter recombination.

D. Impact on observables

We assess the impact of the dark energy sound speed in the CMB and matter power spectra, considering both the PPF and modified gravity scenarios, assuming a dynamical dark energy cosmology with $w_0 = -0.838$ and $w_a = -0.62$, representative of the observed signal [9]. For the modified gravity effects, we consider the parameterization $\alpha_{\rm K} \propto \Omega_{\rm DE}$ with $\lambda_{\rm MG} = 1$ shown in Figure 1.

In Figure 2, we show the impact of varying the dark energy sound speed on the matter power spectrum $P(k, z)$. For the GR case shown in the left panels, lowering the dark energy sound speed impacts the matter power spectrum by less than 1% with respect to the case where $c_s^2 = 1$. Matter perturbations at scales $k > 10^{-3} h/{\rm Mpc}$ are suppressed, while those at scales $k < 10^{-3} h/{\rm Mpc}$ are enhanced. As the the sound speed gets lower, the suppression extends to increasingly smaller scales. The MG case shows an opposite effect: [**Edit**: compared to the GR case, the matter power spectrum is enhanced at small scales, $k > 10^{-3} h/{\rm Mpc}$, by $\approx 5\%$, while suppressed at large scales, $k < 10^{-3} h/{\rm Mpc}$, by $10 - 20\%$ depending on redshift and c_s^2. Both the large-scale power suppression and small-scale power enhancement are increased as the sound speed is increased, although the small scales are less sensitive to the c_s^2 parameter.]

The impact of lowering the sound speed on the CMB anisotropies is similar, as shown in Figure 3. In GR, the temperature power spectrum is enhanced at large scales dominated by cosmic variance, varying by at most 2.5%, while the lensing spectrum is suppressed by the same amount. In the MG case, the effect on the temperature power spectrum is larger, suppressing multipoles $\ell < 100$ by around $20\% - 30\%$. In the lensing power spectrum, we find an overall $\lesssim 5\%$ increase at multipoles $\ell > 100$, which can become as large as 25% at small multipoles.

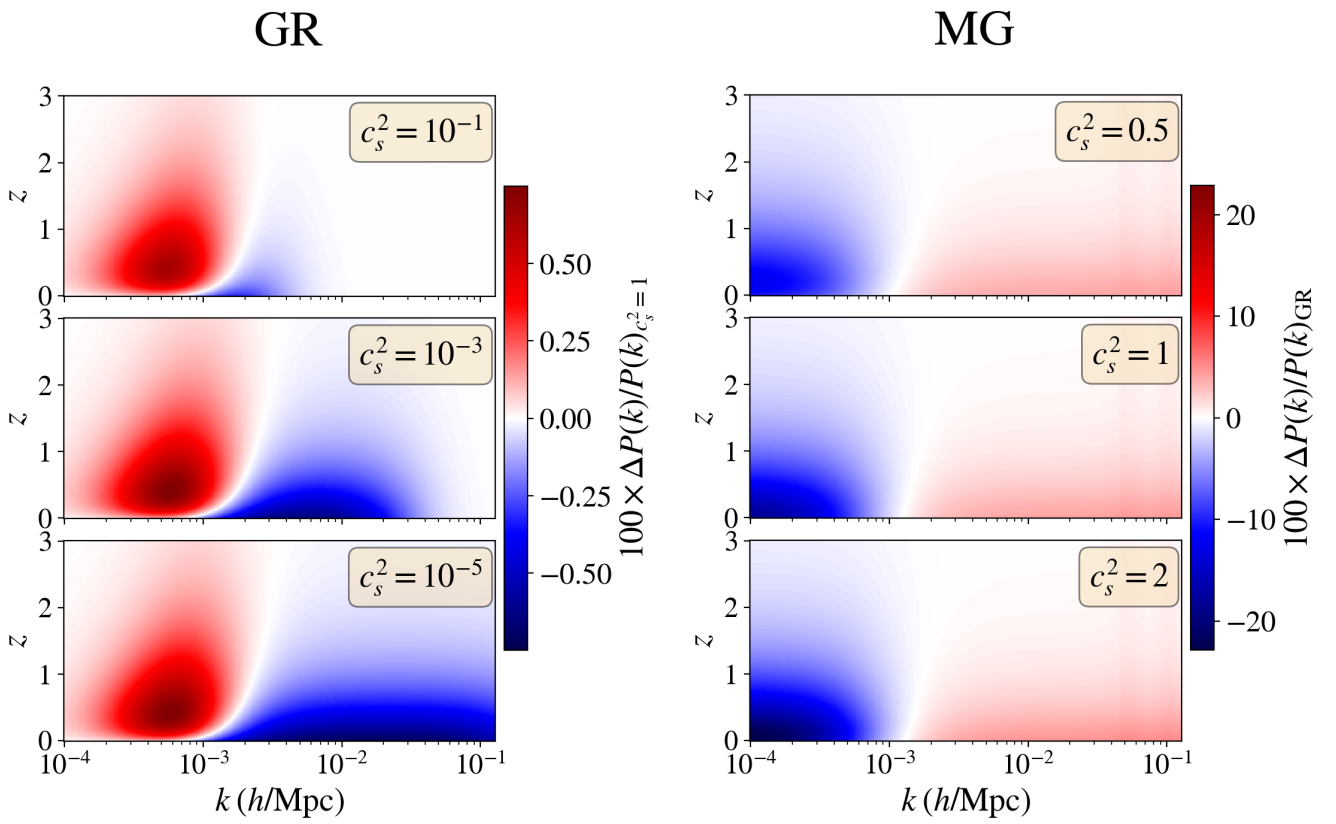


FIG. 2. Relative changes in the matter power spectrum for PPF dark energy in GR (left panels) and MG with $\alpha_{\rm K} \propto \Omega_{\rm DE}$ (right panels), compared to the GR case where $c_s^2 = 1$. Here, we assume $w_0 = -0.838$ and $w_a = -0.67$.

Therefore, as already assessed by previous works [66–70, 88], the small impact of the dark energy sound speed in GR hinders its detectability with current and future observations. However, in MG models, the relation between $w_{\rm DE}$, c_s^2, and the MG functions μ and Σ has the potential to provide useful constraints, as we investigate in the following.

III. DATA ANALYSIS

A. Datasets

1. *Planck PR4 CMB*

We use data from Planck PR4, implemented in the `HiLLiPoP` and `LoLLiPoP` likelihoods [2]. The `HiLLiPoP` likelihood contains TT, TE and EE angular power spectra from multipoles $\ell = 30$ up to $\ell = 2500$ (for TT) and $\ell = 2000$ (for TE and EE). The `LoLLiPoP` likelihood contains EE angular power spectra for multipoles ranging from $\ell \in [2, 29]$. For the large-scale TT power spectrum, we use the `Commander` likelihood. We additionally include CMB lensing from Planck PR4 `NPIPE` likelihood [89] as well as the cross-correlation between lensing and the integrated Sachs-Wolfe effect (ISWL, [90]).

In the following, the CMB primary spectra dataset will be referred to simply as CMB, and the combination of primary spectra, CMB lensing and ISWL will be denoted as CMBL.

2. *DESI DR2 BAO*

The DESI collaboration has measured the Baryon Acoustic Oscillation scale relative to the baryon drag epoch for several tracers of the large-scale structure, such as galaxies, quasars and the Lyman-α forest. The effective redshifts of the tracers samples span the redshift range $0.30 < z < 2.33$. In the following, this dataset will be simply referred to as BAO.

3. *DES-Dovekie*

We use luminosity distances from ≈ 1600 type Ia supernovae catalogued by the DES collaboration in 5 years of observations. In particular, we use the updated magnitudes published by DES-Dovekie, a recalibration of the photometric redshifts of the DES-Y5 supernovae catalog [20]. While this recalibration has reduced the statistical significance of preference for dynamical dark en-

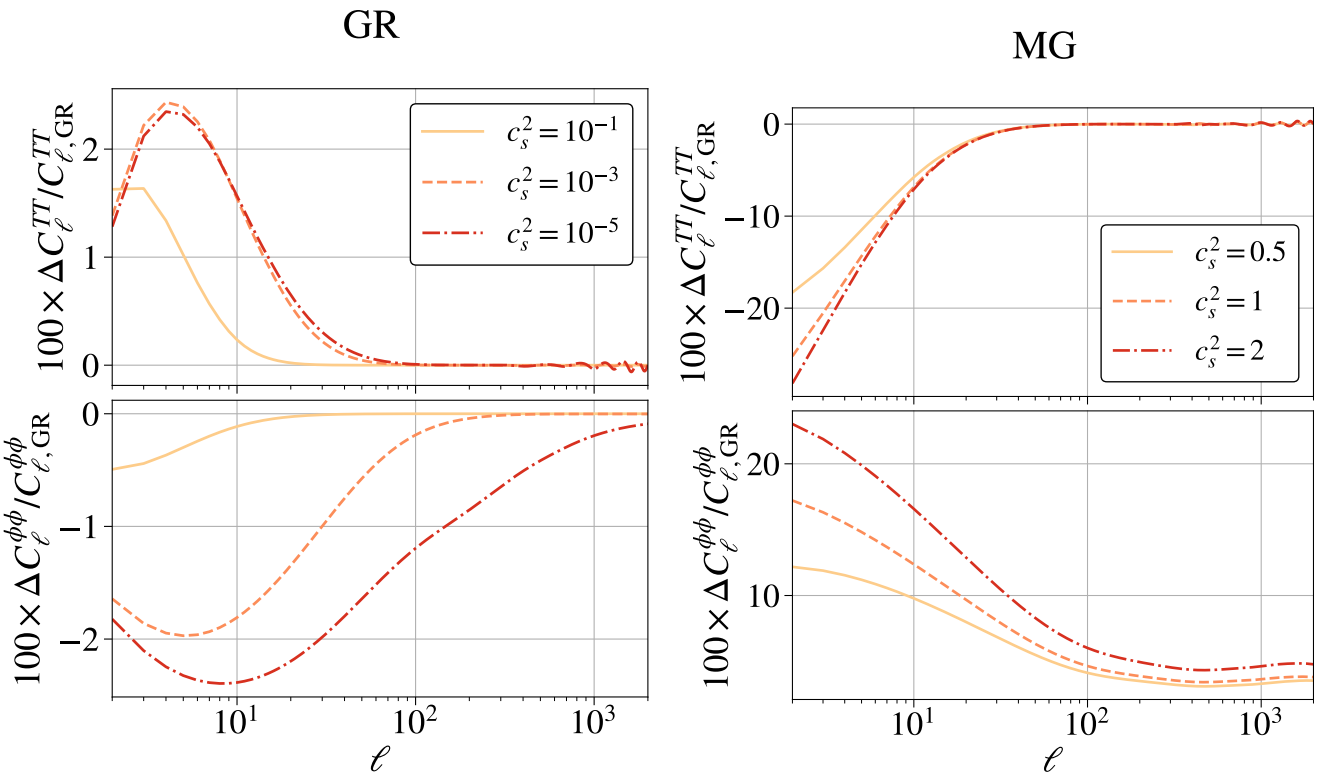


FIG. 3. Relative changes in the CMB anisotropy power spectra for PPF dark energy in GR (left panels) and and MG with $\alpha_{\rm K} \propto \Omega_{\rm DE}$ (right panels), compared with the GR case where $c_s^2 = 1$. We show the effects on the TT (top panels) and $\phi\phi$ (bottom panels) spectra.

ergy from 4.2σ to 3.2σ, it also improved the agreement with other supernovae datasets such as Pantheon+ and Union3.1 [17, 22]. This dataset will be referred to as SN.

4. *DES-Y3 Cosmic Shear*

We use cosmic shear real-space angular correlation functions $\xi_\pm(\theta)$ calculated from the Dark Energy Survey Year 3 dataset [91, 92]. In the following, this dataset will be referred to as CS. The correlations are measured from the METACALIBRATION shape catalog [93], consisting of approximately 100 million galaxies, with an effective number density of [**Edit**: $n_{\rm eff} = 5.59$ arcmin^{-2}]. The galaxies in this catalog are divided in 4 tomographic redshift bins with number density per redshift $n^i(z)$. The angular correlation functions are measured in 20 logarithmically-spaced angular bins in the range $2.5' < \theta < 250'$.

To make theoretical predictions for $\xi_\pm$, we use the `CosmoLike` code [94]. After calling `CAMB`, we can obtain the matter power spectrum $P_m(k, z)$, the comoving distances $\chi(z)$, and the MG functions $\mu(z) = \Sigma(z)$. With that information, we can compute the Fourier-space correlations using the Limber approximation,

$$C^{ij}_{\kappa\kappa,\ell} = \int_0^{\chi_{\rm H}} d\chi \frac{q^i_\kappa(\chi) q^j_\kappa(\chi)}{\chi^2} \Sigma^2(\chi) P_m\left(\frac{\ell + 1/2}{\chi}, z(\chi)\right), \tag{20}$$

where $\chi_{\rm H}$ is the comoving horizon and $q^i_\kappa(\chi)$ is the lensing efficiency for the sources in tomographic bin i,

$$q^i_\kappa(\chi) = \frac{3H_0^2\Omega_m}{2} \frac{\chi}{a(\chi)} \int_\chi^{\chi_{\rm H}} d\chi' \frac{\chi' - \chi}{\chi'} n^i(z(\chi')) \frac{dz}{d\chi'}. \tag{21}$$

Note that the Weyl potential $\Phi + \Psi$ and matter density are related by Equation 16b, thus warranting an extra factor of $\Sigma^2(\chi)$ in Equation 20 compared to the GR case.

With the harmonic space angular power spectrum, we find the real space power spectrum by convolving with Legendre polynomials,

$$\xi^{ij}_\pm(\theta) = \sum_\ell \frac{2\ell + 1}{4\pi} C^{ij}_{\kappa\kappa,\ell} G^\pm_{\ell,2}(\cos\theta), \tag{22}$$

where $G^\pm_{\ell,2}$ are defined in *e.g.* Ref. [95].

The theoretical prediction is modified to account for several systematic effects:

- Photometric redshift uncertainties: to account for uncertainties in the photometric redshift calibration, we distort the galaxy number densities with a

shift parameter Δz^i such that $n^i(z) \to n^i(z+\Delta z^i)$. The shift parameters, one for each tomographic bin, are sampled in the MCMC.

- Multiplicative shear bias: the measurement of the shear from the galaxy catalog is subject to systematic errors. To account for these errors, we scale our shear predictions as $\xi_\pm^{ij} \to (1+m^i)(1+m^j)\xi_\pm^{ij}$ and marginalize over the parameters m^i.

- Intrinsic alignment (IA): unlensed galaxies' shapes can be correlated due to local density and tidal fields. This accounts for extra contributions in the harmonic space shear power spectrum: $C^{ij}_{\kappa\kappa,\ell} \to C^{ij}_{\kappa\kappa,\ell} + C^{ij}_{\kappa I,\ell} + C^{ij}_{I\kappa,\ell} + C^{ij}_{II,\ell}$, where $C^{ij}_{\kappa I,\ell}$ are the correlations between cosmic shear at bin i and IA at bin j and $C^{ij}_{II,\ell}$ is the IA autocorrelation. We use the Nonlinear Linear Alignment (NLA) model, where the intrinsic shape field is sourced by the gravitational tidal field. Thus, the extra correlations are given by

$$C^{ij}_{\kappa I,\ell} = \int_0^{\chi_H} d\chi \, \frac{n^i(\chi) q^j(\chi)}{\chi^2} A_{\rm IA}(\chi)\Sigma(\chi)\mu(\chi) \times P_m\left(\frac{\ell+1/2}{\chi}, z(\chi)\right), \tag{23a}$$

$$C^{ij}_{II,\ell} = \int_0^{\chi_H} d\chi \, \frac{n^i(\chi) n^j(\chi)}{\chi^2} A^2_{\rm IA}(\chi)\mu^2(\chi) \times P_m\left(\frac{\ell+1/2}{\chi}, z(\chi)\right). \tag{23b}$$

  where $A_{\rm IA}$ is a redshift-dependent amplitude of the intrinsic alignment effect. Again, note that the gravitational potential Ψ and matter density are related by Equation 16a, and thus we include an additional $\mu(\chi)$ factor when relating the tidal field $s_{ij} = (k_i k_j/k^2 - \delta_{ij}/3)\Psi$ to the matter density [96] We model the time-dependence of the IA amplitude as a power law given by

$$A_{\rm IA}(z) = A_1 C_1 \rho_{\rm cr} \left(\frac{1+z}{1+z_p}\right)^{\eta_1}, \tag{24}$$

  where A_1 and η_1 are parameters characterizing the IA redshift evolution, $C_1 = 5\times10^{-14} M_\odot h^{-2}{\rm Mpc}^2$ is a conventional constant to scale the amplitude based on IA measurements from the SuperCOSMOS survey [97], $\rho_{\rm cr}$ is the critial energy density today, and z_p is an arbitrary pivot redshift.

Priors for the nuisance parameters Δz^i, m^i, A_1 and η_1 are shown in Table I.

B. Modelling

We assume dark energy is a cosmological fluid whose background dynamics is defined by an equation of state parameterized by

$$w_{\rm DE}(a) = w_0 + w_a(1-a). \tag{25}$$

For the perturbative dynamics, we investigate the following scenarios involving the sound speed c_s^2:

- Fiducial: we use the PPF equations for dark energy perturbations assuming $c_s^2 = 1$.

- Modified Gravity: we solve Equation 9, finding $\alpha_{\rm B}$ and μ, and explicitly changing the metric equations according to Equations 16. We assume the four parameterizations for $\alpha_{\rm K}$ described in Section II B. Our standard choice is to assume a constant, subluminal sound speed $c_s^2 < 1$. We also test the effect of a superluminal sound speed, where we allow $c_s^2 < 10$, and a dynamical sound speed, where $c_s^2 = c_{s,0}^2 + c_{s,a}^2(1-a)$.

We do not use any nonlinear matter power spectrum prescription in our analysis. For DES-Y3 cosmic shear, we make use of linear scale cuts following the methodology of [86, 87]. Furthermore, the CMB primary spectra are mostly insensitive to nonlinear effects. For CMB lensing, while neglecting nonlinear effects can lead to small biases in ΛCDM parameters, erroneously using the nonlinear corrections from `HMCode2020` [98] can lead to numerical artifacts in the power spectrum prediction [99]. We have performed our analysis with both cases and confirmed that the cosmological constraints agree, implying that our results have limited sensitivity to nonlinear effects. However, we have noticed that MCMCs ran using `HMCode2020` have been plagued with stuck walkers and slow convergence attributed to the erroneous nonlinear prescription.

C. Analysis

We use `Cocoa`[2], the Cobaya-Cosmolike Joint Architecture [94], as our framework for cosmological analysis. For sampling the posterior distributions, we use the Metropolis-Hastings algorithm implemented in `Cobaya`, using the Gelman-Rubin criterion with $|R-1| < 0.03$ to assess convergence. We sample over the standard cosmological parameters $\{\theta_*, \Omega_b h^2, \Omega_c h^2, \ln(10^{10}A_s), n_s, \tau\}$, as well as the dark energy parameters $\{w_0, w_a, c_s^2\}$. For the modified gravity scenarios where $\alpha_{\rm K}$ is parameterized, we additionally sample over $\lambda_{\rm MG}$. The cosmological parameter priors are described in Table I. We also sample over the CMB and supernovae nuisance parameters following the collaborations' recommendations. We consider two dataset combinations: CMB+BAO+SN, and CMBL+BAO+SN+CS.

[2] https://github.com/CosmoLike/cocoa

Cosmology		DES-Y3 Nuisance	
Parameter	Prior	Parameter	Prior
$100 \times \theta_*$	$\mathcal{U}[0.5, 10]$	Δz^1	$\mathcal{N}[0, 0.018]$
$\Omega_b h^2$	$\mathcal{U}[0.005, 0.1]$	Δz^2	$\mathcal{N}[0, 0.015]$
$\Omega_c h^2$	$\mathcal{U}[0.001, 0.99]$	Δz^3	$\mathcal{N}[0, 0.011]$
$\ln(10^{10} A_s)$	$\mathcal{U}[1.61, 3.91]$	Δz^4	$\mathcal{N}[0, 0.017]$
n_s	$\mathcal{U}[0.8, 1.2]$	$100 \times m^1$	$\mathcal{U}[-0.6, 0.9]$
τ	$\mathcal{U}[0.01, 0.8]$	$100 \times m^2$	$\mathcal{U}[-2.0, 0.8]$
w_0	$\mathcal{U}[-3(-1), -1/3]$	$100 \times m^3$	$\mathcal{U}[-2.4, 0.8]$
$w_0 + w_a$	$\mathcal{U}[-3(-1), 1]$	$100 \times m^4$	$\mathcal{U}[-3.7, 0.8]$
c_s^2	$\mathcal{U}[0, 1(10)]$	A_1	$\mathcal{U}[-5, 5]$
$c_{s,0}^2 + c_{s,a}^2$	$\mathcal{U}[0, 1(10)]$	η_1	$\mathcal{U}[-5, 5]$
$\lambda_{\rm MG}$	$\mathcal{U}[0, 3]$	–	

TABLE I. Priors for cosmological and nuisance parameters sampled in the MCMCs for different modelling choices. Values in parenthesis denote alternative prior choices tested in this work.

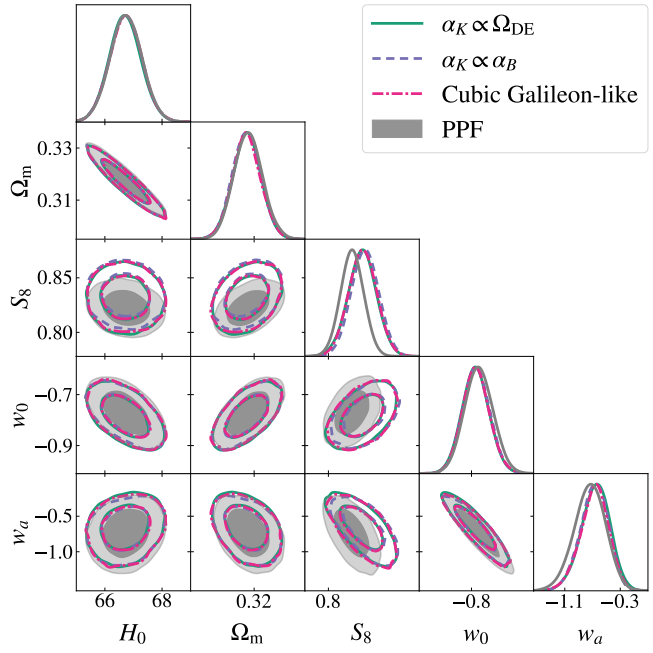


FIG. 4. 2D posterior contours (68% and 95% credible regions) for cosmological parameters, assuming the dataset combination of CMB+BAO+SN. The gray contour shows the GR case and the colored contours show the MG cases with three different parameterizations for $\alpha_{\rm K}$: proportional to $\Omega_{\rm DE}$ (green, solid), proportional to $\alpha_{\rm B}$ (purple, dashed), and the cubic Galileon-like (pink, dotted-dashed).

IV. RESULTS AND DISCUSSION

We begin by discussing cosmological parameter constraints for the CMB+BAO+SN dataset combination. Figure 4 shows constraints on the standard cosmological parameters, as well as on w_0 and w_a. Results for the k-essence-like parameterization are shown in Appendix A, since this case requires the additional priors $w_0 > -1$ and $w_0 + w_a > -1$. The marginalized constraints, reported as means and 68% confidence intervals, are shown in Table II.

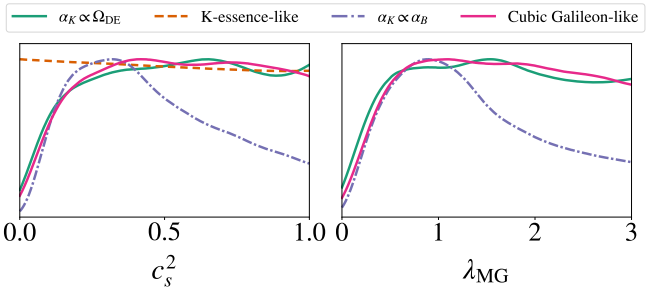


FIG. 5. 1D marginalized constraints on the perturbative parameters c_s^2 and $\lambda_{\rm MG}$ assuming the dataset combination of CMB+BAO+SN.

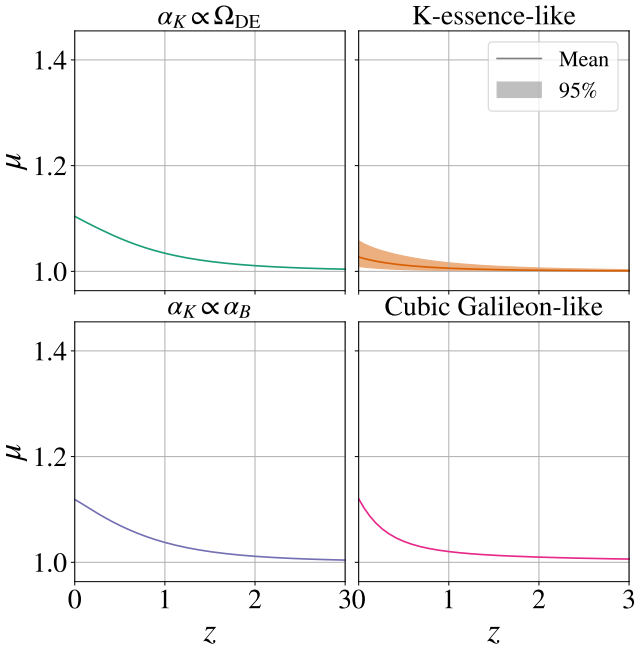


FIG. 6. Constraints on $\mu(a)$ for the combination of CMB+BAO+SN, for all parameterizations of $\alpha_{\rm K}$ considered. The thick lines shows the means while the bands show the 95% confidence intervals.

We first note that the cosmological parameter constraints in the MG cases agree with one another, showing that the three different parametrizations have similar qualitative behavior with respect to the μ and Σ functions. The MG and GR cases yield consistent constraints on H_0 and Ω_m. The S_8 constraints obtained in MG are slightly shifted towards higher values because $\mu \geq 1$ in our MG models, enhancing the formation of structure. The marginalized constraints, reported as means and 68% confidence intervals, are

- $S_8 = 0.822 \pm 0.011$ (GR),
- $S_8 = 0.832 \pm 0.013$ (MG, $\alpha_{\rm K} \propto \Omega_{\rm DE}$),

- $S_8 = 0.833 \pm 0.013$ (MG, $\alpha_{\rm K} \propto \alpha_{\rm B}$),
- $S_8 = 0.832 \pm 0.013$ (MG, Cubic Galileon-like).

Interestingly, due to the direct influence of the dark energy equation of state on the growth of structure, the constraints on w_0 and w_a are slightly tighter in the MG case than in the GR case. The dark energy figure of merit, defined as the reciprocal of the area of the 95% confidence region in the $w_0 - w_a$ posterior [100], is increased by approximately 20% in the MG models. We find 68% confidence intervals of

- $w_0 = -0.77 \pm 0.06$, $w_a = -0.74^{+0.24}_{-0.22}$ (GR),
- $w_0 = -0.78 \pm 0.06$, $w_a = -0.66^{+0.22}_{-0.19}$ (MG, $\alpha_{\rm K} \propto \Omega_{\rm DE}$),
- $w_0 = -0.78 \pm 0.05$, $w_a = -0.66^{+0.20}_{-0.17}$ (MG, $\alpha_{\rm K} \propto \alpha_{\rm B}$),
- $w_0 = -0.78 \pm 0.06$, $w_a = -0.67^{+0.22}_{-0.19}$ (MG, Cubic Galileon-like).

Figure 5 shows marginalized constraints on c_s^2 and $\lambda_{\rm MG}$. The c_s^2 posteriors are mostly flat, indicating weak constraining power. As shown in Figure 3, the impact of the MG models on the CMB is concentrated at low multipoles dominated by cosmic variance. In all cases except the k-essence-like model, the posterior is mostly flat across the prior, but disfavors low values of c_s^2 that tend to increase μ. These cases, however, remain consistent with $c_s^2 = 0$ at 95% confidence level. Finally, the c_s^2 posterior is completely flat for the k-essence-like case. Similar remarks apply to the posterior distribution of the phenomenological $\lambda_{\rm MG}$ parameter. Interestingly, for the case where $\alpha_{\rm K} \propto \alpha_{\rm B}$, purple dot-dashed curves, the 1D posterior is consistent with $\lambda_{\rm MG} = 1$, which is the limit where this parametrization recovers the relation $\alpha_{\rm K} = 3\alpha_{\rm B}$ from the Cubic Galileon model [83].

In Figure 6, we show constraints (mean and 95% limits) on $\mu(a)$ for the four parameterizations of $\alpha_{\rm K}$. Interestingly, for all models, we find a deviation from $\mu = 1$ at over 2σ confidence level at redshifts $z < 2$. Except for the k-essence-like model, which has non-phantom priors on w_0 and w_a, all phenomenological parameterizations agree very well with one another, with present-day mean values $\mu_0 \coloneqq \mu(z = 0) \approx 1.1$; the k-essence-like parameterization instead gives $\mu_0 \approx 1.04$.

We interpret the evidence for $\mu > 1$ as being driven by the preference for a phantom dynamical dark energy rather than by an improvement in the CMB fit. First, we remark that our theory space is mostly restricted to $\mu \geq 1$. The growth of $\alpha_{\rm B}$ is driven by dark energy at low redshifts when $\Omega_{\rm DE}(a)$ is non-negligible and $w_{\rm DE} \neq -1$. Since the CMB+BAO+SN data combination has a significant preference for dynamical dark energy, this drives the late-time evolution of $\alpha_{\rm B}$, leading to deviations from GR. This also explains why the k-essence-like parameterization shows a weaker deviation from $\mu = 1$: because of its restriction to $w_{\rm DE} \geq -1$, its constraints are compatible with ΛCDM within the 2σ confidence level, and dark energy remains constant until thawing at $z \approx 0.5$. To test this hypothesis, we first compare the distribution of $\chi^2_{\rm CMB}$ between the GR and MG cases, using $\alpha_{\rm K} \propto \Omega_{\rm DE}$ as a representative model. Figure 7 shows the distribution of $\chi^2_{\rm CMB}$ in both cases. While the MG $\chi^2_{\rm CMB}$ is shifted towards lower $\chi^2_{\rm CMB}$ compared to the GR case, this is expected due to the inclusion of two extra parameters, c_s^2 and $\lambda_{\rm MG}$. The shift of $\Delta\chi^2_{\rm CMB} \approx 1$ is small relative to the additional degrees of freedom in the MG model. We conclude that the MG effects do not produce any significant improvement in the CMB goodness-of-fit.

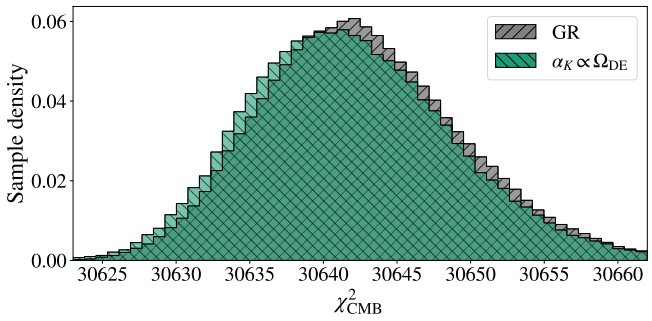


FIG. 7. Distribution of $\chi^2_{\rm CMB}$ in the MCMC samples. The [**Edit**: green histogram shows one representative MG case, and the gray] histogram shows the GR case.

To further test our hypothesis, we repeat our analysis while fixing the equation of state to -1, *i.e.* setting $w_0 = -1$ and $w_a = 0$. The constraints on μ for this case are shown in the left panel of Figure 8. As stated above, for the k-essence-like and $\alpha_{\rm K} \propto \alpha_{\rm B}$ parameterizations, a cosmological constant implies $\alpha_{\rm B} = 0$ at all times, and therefore $\mu = 1$. For the Cubic Galileon-like and $\alpha_{\rm K} \propto \Omega_{\rm DE}$ parameterizations, the evolution of μ is still observed, but the constraints are now compatible with GR at the 2σ confidence level. This result highlights the interplay between the background expansion history and modified gravity constraints in theoretical models.

We have confirmed that our results remain consistent when adopting a superluminal prior of $c_s^2 < 10$. As discussed above, higher values of c_s^2 decrease the maximum value of μ, bringing the model closer to the GR phenomenology. The middle panel of Figure 8 shows constraints on $\mu(a)$ under this prior. While the mean value of $\mu(a = 1)$ drops slightly below 1.1, the conclusions drawn from the subluminal prior case still hold.

Furthermore, we have tested the effect of a dynamical sound speed on our results. We find that the coefficient $c^2_{s,a}$ is compatible with zero in all cases. An important consequence of this, is that while we expect that models with a dynamical scalar field have a time evolving sound speed, this feature is not detected by the data. Therefore, treating c_s^2 as a constant parameter is sufficient in the context of the datasets we consider in this work. The right panel of Figure 8 shows the constraints on $\mu(a)$

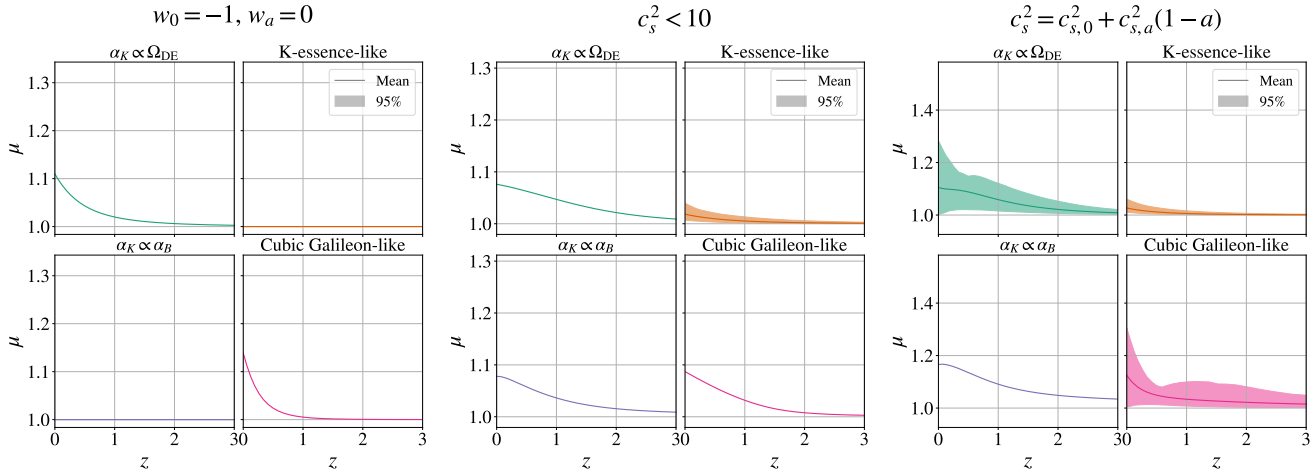


FIG. 8. Same as Figure 6, but with different priors on cosmological parameters: the left panel assumes a cosmological constant, the middle panel assumes a prior of $0 < c_s^2 < 10$, and the right panel assumes a dynamical sound speed of the form $c_s^2 = c_{s,0}^2 + c_{s,a}^2(1-a)$.

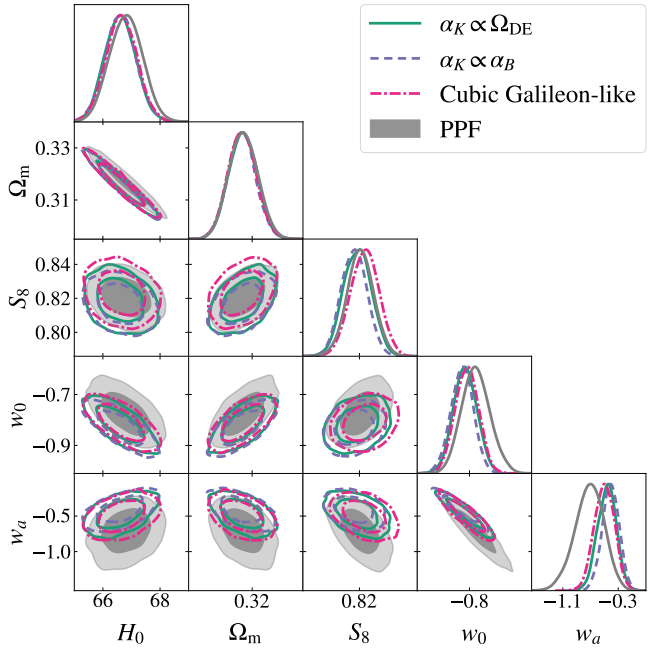


FIG. 9. Same as Figure 4, but for the CMBL+BAO+SN+CS dataset combination.

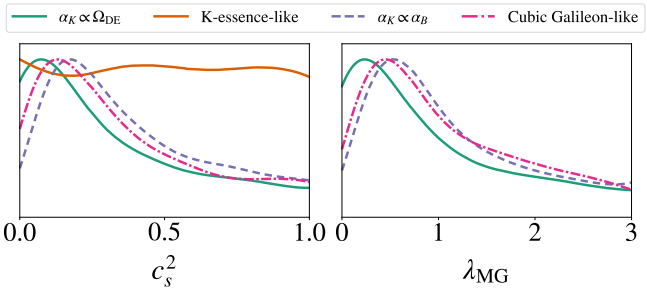


FIG. 10. Same as Figure 5, but for the CMBL+BAO+SN+CS dataset combination.

assuming a dynamical sound speed. While these constraints remain mostly unchanged, the 95% upper bound on $\mu(a)$ increases significantly in the $\alpha_{\rm K} \propto \alpha_{\rm B}$ case. We have confirmed that these deviations occur only at redshifts $z \leq 20$.

We now turn our attention to the constraints for the CMBL+BAO+SN+CS dataset combination. Figure 9 shows confidence contours for the cosmological parameters, Figure 10 shows the marginalized one-dimensional constraints on c_s^2 and $\lambda_{\rm MG}$, Figure 11 shows constraints on $\mu(a)$ and Table II shows 1D marginalized constraints.

Unlike the case without cosmic shear and CMB lensing, the constraints on S_8 obtained from the GR and MG cases now agree with one another. Furthermore, the posteriors of w_0 and w_a are significantly shifted towards ΛCDM in the MG cases. The one-dimensional marginalized constraints, reported as means and 68% confidence intervals, are:

- $w_0 = -0.78 \pm 0.07,\ w_a = -0.72^{+0.22}_{-0.20}$ (GR),
- $w_0 = -0.82 \pm 0.05,\ w_a = -0.46 \pm 0.14$ (MG, $\alpha_{\rm K} \propto \Omega_{\rm DE}$),
- $w_0 = -0.83 \pm 0.05,\ w_a = -0.41^{+0.13}_{-0.11}$ (MG, $\alpha_{\rm K} \propto \alpha_{\rm B}$),
- $w_0 = -0.81 \pm 0.05,\ w_a = -0.50 \pm 0.15$ (MG, Cubic Galileon-like).

These results suggest a weaker deviation from GR than in the case without LSS probes. Indeed, from Figure 11 and Table II, we see that the mean value of μ_0 decreases from 1.1 to 1.02 $\alpha_{\rm K} \propto \Omega_{\rm DE}$ and Cubic Galileon-like cases, and to 0.99 in the $\alpha_{\rm K} \propto \alpha_{\rm B}$. These values represent

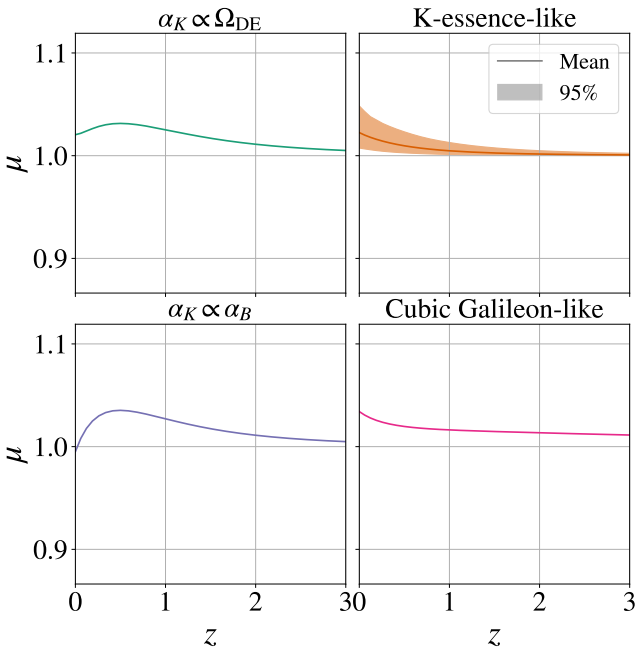


FIG. 11. Same as Figure 6, but for the CMBL+BAO+SN+CS dataset combination.

a much weaker modification of gravity than in the case without DES-Y3 and CMB lensing. The k-essence-like case is practically unchanged. We conclude that large-scale structure probes are in tension with the previous results of $\mu > 1$, driving the resulting constraints towards GR. Due to the correlations between μ and the dark energy equation of state, the constraints on the latter are shifted towards a cosmological constant.

Even with additional information from large-scale structure, the constraints on the dark energy sound speed, shown in Figure 10, remain weak. For the cases where $w_{\rm DE}$ is allowed to be phantom, the posteriors peak at values around $c_s^2 \approx 0.2$, but are still compatible with the prior boundaries at 95% confidence level. The constraints on $\lambda_{\rm MG}$ peak at values slightly below one and weakly disfavor higher values.

A. Quasi-Static Approximation

V. CONCLUSIONS

Hints for dynamical dark energy from current surveys have motivated the investigation of new theoretical models that can consistently explain the observed signal in the $w_0 w_a$ parameter space. These models, however, may differ substantially in their predictions for the growth of matter inhomogeneities. Future experiments, including galaxy surveys, CMB measurements, and 21-cm line intensity mapping, will provide a wealth of information about large-scale structure and will therefore be able to distinguish between candidate models for dark energy. In this context, the dark energy sound speed, usually assumed to be unity, provides a physically motivated parameter for testing the consistency of the $w_0 w_a$ expansion history and growth of matter perturbations. While in GR the effects of a non-standard c_s^2 are small and concentrated at large scales, in modified gravity both the dark energy sound speed and equation of state are directly tied to changes in matter clustering at small scales.

In this work, we have investigated the direct relation between dark energy properties, including its equation of state $w_{\rm DE}$ and sound speed c_s^2, and the μ and Σ functions that modify the growth of structure and light propagation. By using c_s^2 as an input parameter instead of the MG functions α_i, we are able to solve for $\mu(a)$ and $\Sigma(a)$, provided additional parametrizations for $\alpha_{\rm K}$ are specified. We have tested four parametrizations, finding consistent results among the three of them that allow phantom crossing. This framework has the additional advantage of ensuring that the dark energy perturbations are stable, with $c_s^2 > 0$ and $D_{\rm kin} > 0$.

We find that the direct impact of the dark energy equation of state on matter perturbations increases the constraining power on w_0 and w_a obtained from CMB and LSS probes. For the CMB+BAO+SN dataset combination, we find a preference for $\mu > 1$ at low redshifts at over the 2σ significance, consistent with previous results in the literature [101]. This signal is driven by the preference for phantom dark energy rather than by an improvement in the CMB fit. Furthermore, this preference persists if we assume a superluminal sound speed or a dynamical sound speed, but vanishes if we assume a ΛCDM background. This result highlights the correlations between background physics and the growth of structure, especially in modified gravity models or in dark energy models with an imperfect stress-energy tensor. When including information from cosmic shear and CMB lensing, we find that the strength of the deviations from GR significantly decreases, along with a shift in w_0 and w_a towards a cosmological constant.

Overall, this theoretical framework can simultaneously test the consistency of the ΛCDM model at both background and perturbative levels. Our results can be further extended by including a model for the $\alpha_{\rm M}$ function, which controls the dark energy anisotropic stress. This framework can also be readily implemented in N-body codes [71], relaxing the need for the linear scale cuts used for cosmic shear in this work. In particular, the Comoving Lagrangian Acceleration (COLA) algorithm [102] has proven suitable for generating large simulation suites for machine learning models to learn the nonlinear corrections to the matter power spectrum [103, 104]. In the future, with a suitable model for nonlinear gravitational effects, we aim to further extend our results by forecasting constraints for future galaxy surveys.

	w_0w_aCDM	$\alpha_K \propto \Omega_{DE}$	$\alpha_K \propto \alpha_B$	Cubic Galileon-like	K-essence-like	w_0w_aCDM $w \geq -1$
	CMB+BAO+SN					
Ω_m	0.318 ± 0.006	0.317 ± 0.006	0.317 ± 0.005	0.317 ± 0.006	0.312 ± 0.005	0.312 ± 0.006
H_0	66.7 ± 0.6	66.7 ± 0.6	66.7 ± 0.5	66.7 ± 0.6	66.8 ± 0.5	66.9 ± 0.6
S_8	0.822 ± 0.011	0.832 ± 0.013	0.833 ± 0.013	0.832 ± 0.013	0.795 ± 0.010	0.798 ± 0.010
w_0	-0.77 ± 0.06	-0.78 ± 0.06	-0.78 ± 0.05	-0.78 ± 0.06	-0.93 ± 0.03	-0.93 ± 0.03
w_a	$-0.74^{+0.24}_{-0.22}$	$-0.66^{+0.22}_{-0.19}$	$-0.66^{+0.20}_{-0.17}$	$-0.67^{+0.22}_{-0.19}$	$-0.03^{+0.03}_{-0.06}$	$-0.04^{+0.04}_{-0.05}$
μ_0	1	$1.10^{+0.09}_{-0.08}$	$1.11^{+0.06}_{-0.08}$	$1.12^{+0.10}_{-0.09}$	1.03 ± 0.01	1
	CMBL+BAO+SN+CS					
Ω_m	0.316 ± 0.006	0.316 ± 0.006	0.316 ± 0.005	0.316 ± 0.006	0.311 ± 0.005	0.312 ± 0.005
H_0	66.8 ± 0.7	66.6 ± 0.6	66.6 ± 0.5	66.6 ± 0.6	67.0 ± 0.5	67.0 ± 0.6
S_8	0.821 ± 0.008	0.819 ± 0.008	0.817 ± 0.008	0.823 ± 0.009	0.803 ± 0.008	0.804 ± 0.007
w_0	-0.78 ± 0.07	-0.82 ± 0.05	-0.83 ± 0.05	-0.81 ± 0.05	-0.94 ± 0.03	-0.94 ± 0.03
w_a	$-0.72^{+0.22}_{-0.20}$	-0.46 ± 0.14	$-0.41^{+0.13}_{-0.11}$	-0.50 ± 0.15	$-0.02^{+0.04}_{-0.05}$	$-0.04^{+0.04}_{-0.05}$
μ_0	1	$1.02^{+0.02}_{-0.02}$	$0.99^{+0.04}_{-0.04}$	$1.03^{+0.03}_{-0.03}$	1.02 ± 0.01	1

TABLE II. Constraints (mean and 68% confidence intervals) on the cosmological parameters Ω_m, H_0, S_8, w_0, w_a as well as μ_0, the present value of the MG function $\mu(a)$ (see Equation 16).

ACKNOWLEDGMENTS

The authors thank Kazuya Koyama for useful comments. This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) – Finance Code 001. JR acknowledges financial support from CAPES. FTF acknowledges financial support from the National Scientific and Technological Research Council (CNPq, Brazil). This work made use of the CHE cluster, managed and funded by COSMO/CBPF/MCTI, with financial support from CNPq, FINEP and FAPERJ.

DATA AVAILABILITY

The `CAMB-cs2` code used in this work is publicly available at https://github.com/joaoreboucas1/CAMB-cs2. MCMC samples are available from the authors upon reasonable request.

Appendix A: Results for K-essence-like Parameterization

In this Appendix, we show posterior contour plots for the k-essence-like parametrization, which were omitted from the main text for the sake of conciseness. Figure 12 shows the posterior confidence contours for the CMB+BAO+SN and CMBL+BAO+SN+CS dataset combinations.

[1] Planck Collaboration, N. Aghanim, Y. Akrami, M. Ashdown, J. Aumont, *et al.*, Planck 2018 results. VI. Cosmological parameters, Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209 [astro-ph.CO].

[2] M. Tristram, A. J. Banday, M. Douspis, X. Garrido, K. M. Górski, *et al.*, Cosmological parameters derived from the final Planck data release (PR4), Astronomy & Astrophysics **682**, A37 (2024), arXiv:2309.10034 [astro-ph.CO].

[3] S. Aiola, E. Calabrese, L. Maurin, S. Naess, B. L. Schmitt, *et al.*, The Atacama Cosmology Telescope: DR4 maps and cosmological parameters, Journal of Cosmology and Astroparticle Physics **2020** (12), 047, arXiv:2007.07288 [astro-ph.CO].

[4] T. Louis, A. La Posta, Z. Atkins, H. T. Jense, I. Abril-Cabezas, *et al.*, The Atacama Cosmology Telescope: DR6 power spectra, likelihoods and ΛCDM parameters, Journal of Cosmology and Astroparticle Physics **2025**, 062 (2025), arXiv:2503.14452 [astro-ph.CO].

[5] L. Balkenhol, D. Dutcher, A. Spurio Mancini, A. Doussot, K. Benabed, *et al.*, Measurement of the CMB temperature power spectrum and constraints on cosmology from the SPT-3G 2018 T T , T E , and E E dataset, Phys. Rev. D **108**, 023510 (2023), arXiv:2212.05642 [astro-ph.CO].

[6] E. Camphuis, W. Quan, L. Balkenhol, A. R. Khalife, F. Ge, *et al.*, SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G Main field, arXiv e-prints , arXiv:2506.20707 (2025), arXiv:2506.20707 [astro-ph.CO].

[7] B. Stölzner, A. H. Wright, M. Asgari, C. Heymans, H. Hildebrandt, *et al.*, KiDS-Legacy: Consistency of cosmic shear measurements and joint cosmological constraints with external probes, Astronomy & Astrophysics **702**, A169 (2025), arXiv:2503.19442 [astro-ph.CO].

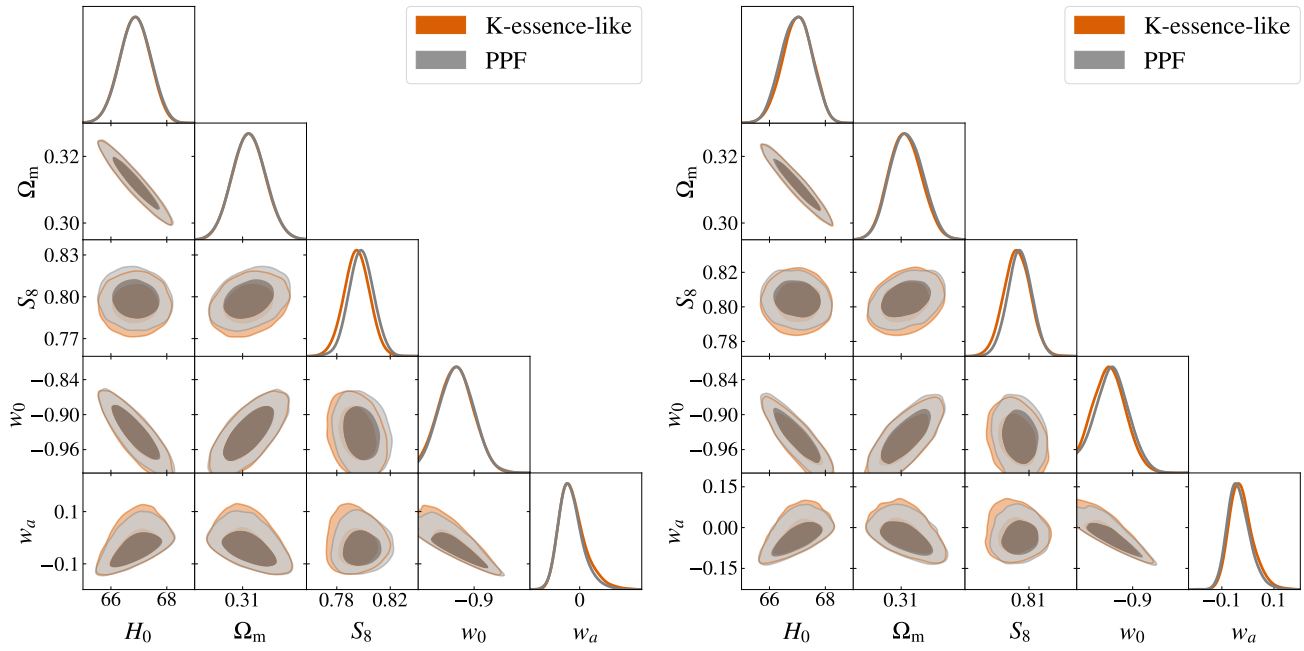


FIG. 12. 2D cosmological parameter posterior contours (68% and 95% credible regions). The left plot shows the posteriors for the dataset combination CMB+BAO+SN, while the right plot shows the posteriors for CMBL+BAO+SN+CS. The gray contours shows the GR case and the orange contours show the k-essence-like MG case.

[8] A. G. Adame, J. Aguilar, S. Ahlen, S. Alam, D. M. Alexander, *et al.*, DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations, Journal of Cosmology and Astroparticle Physics **2025**, 021 (2025), arXiv:2404.03002 [astro-ph.CO].

[9] M. Abdul Karim, J. Aguilar, S. Ahlen, S. Alam, L. Allen, *et al.*, DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints, Phys. Rev. D **112**, 083515 (2025), arXiv:2503.14738 [astro-ph.CO].

[10] A. G. Adame, J. Aguilar, S. Ahlen, S. Alam, D. M. Alexander, *et al.*, DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars, Journal of Cosmology and Astroparticle Physics **2025**, 008 (2025), arXiv:2411.12021 [astro-ph.CO].

[11] T. M. C. Abbott, F. B. Abdalla, A. Alarcon, J. Aleksić, S. Allam, *et al.*, Dark Energy Survey year 1 results: Cosmological constraints from galaxy clustering and weak lensing, Phys. Rev. D **98**, 043526 (2018), arXiv:1708.01530 [astro-ph.CO].

[12] T. M. C. Abbott, M. Aguena, A. Alarcon, S. Allam, O. Alves, *et al.*, Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and weak lensing, Phys. Rev. D **105**, 023520 (2022), arXiv:2105.13549 [astro-ph.CO].

[13] DES Collaboration, T. M. C. Abbott, M. Adamow, M. Aguena, A. Alarcon, *et al.*, Dark Energy Survey Year 6 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing, arXiv e-prints , arXiv:2601.14559 (2026), arXiv:2601.14559 [astro-ph.CO].

[14] M. Asgari, C.-A. Lin, B. Joachimi, B. Giblin, C. Heymans, *et al.*, KiDS-1000 cosmology: Cosmic shear constraints and comparison between two point statistics, Astron. Astrophys. **645**, A104 (2021), arXiv:2007.15633 [astro-ph.CO].

[15] T. Hamana, M. Shirasaki, S. Miyazaki, C. Hikage, M. Oguri, *et al.*, Cosmological constraints from cosmic shear two-point correlation functions with HSC survey first-year data, Publications of the Astronomical Society of Japan **72**, 16 (2020), arXiv:1906.06041 [astro-ph.CO].

[16] D. M. Scolnic, D. O. Jones, A. Rest, Y. C. Pan, R. Chornock, *et al.*, The Complete Light-curve Sample of Spectroscopically Confirmed SNe Ia from Pan-STARRS1 and Cosmological Constraints from the Combined Pantheon Sample, Astrophys. J. **859**, 101 (2018), arXiv:1710.00845 [astro-ph.CO].

[17] D. Scolnic, D. Brout, A. Carr, A. G. Riess, T. M. Davis, *et al.*, The Pantheon+ Analysis: The Full Data Set and Light-curve Release, Astrophys. J. **938**, 113 (2022), arXiv:2112.03863 [astro-ph.CO].

[18] A. Möller, M. Smith, M. Sako, M. Sullivan, M. Vincenzi, *et al.*, The dark energy survey 5-yr photometrically identified type Ia supernovae, Mon. Not. Roy. Astron. Soc. **514**, 5159 (2022), arXiv:2201.11142 [astro-ph.CO].

[19] DES Collaboration, T. M. C. Abbott, M. Acevedo, M. Aguena, A. Alarcon, *et al.*, The Dark Energy Survey: Cosmology Results With ~1500 New High-redshift Type Ia Supernovae Using The Full 5-year Dataset, arXiv e-prints , arXiv:2401.02929 (2024), arXiv:2401.02929 [astro-ph.CO].

[20] B. Popovic, P. Shah, W. D. Kenworthy, R. Kessler, T. M. Davis, *et al.*, The Dark Energy Survey Supernova Program: A Reanalysis Of Cosmology Results And Evidence For Evolving Dark Energy With An Updated Type Ia Supernova Calibration, arXiv e-prints , arXiv:2511.07517 (2025), arXiv:2511.07517

[astro-ph.CO].
[21] D. Rubin, G. Aldering, M. Betoule, A. Fruchter, X. Huang, *et al.*, Union Through UNITY: Cosmology with 2,000 SNe Using a Unified Bayesian Framework, arXiv e-prints , arXiv:2311.12098 (2023), arXiv:2311.12098 [astro-ph.CO].
[22] T. J. Hoyt, D. Rubin, G. Aldering, S. Perlmutter, A. Cuceu, and R. Gupta, Union3.1: Self-consistent Measurements of Host Galaxy Properties for 2000 Type Ia Supernovae, arXiv e-prints , arXiv:2601.19424 (2026), arXiv:2601.19424 [astro-ph.CO].
[23] L. Galbany, T. de Jaeger, A. G. Riess, T. E. Müller-Bravo, S. Dhawan, *et al.*, An updated measurement of the Hubble constant from near-infrared observations of Type Ia supernovae, Astron. Astrophys. **679**, A95 (2023), arXiv:2209.02546 [astro-ph.CO].
[24] H0DN Collaboration, S. Casertano, G. Anand, R. I. Anderson, R. Beaton, *et al.*, The Local Distance Network: a community consensus report on the measurement of the Hubble constant at 1% precision, arXiv e-prints , arXiv:2510.23823 (2025), arXiv:2510.23823 [astro-ph.CO].
[25] E. Abdalla, G. F. Abellán, A. Aboubrahim, A. Agnello, Ö. Akarsu, *et al.*, Cosmology intertwined: A review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies, Journal of High Energy Astrophysics **34**, 49 (2022), arXiv:2203.06142 [astro-ph.CO].
[26] A. Amon and G. Efstathiou, A non-linear solution to the S_8 tension?, Mon. Not. of the Roy. Astron. Soc. **516**, 5355 (2022), arXiv:2206.11794 [astro-ph.CO].
[27] I. G. McCarthy, J. Salcido, J. Schaye, J. Kwan, W. Elbers, *et al.*, The FLAMINGO project: revisiting the S_8 tension and the role of baryonic physics, Mon. Not. of the Roy. Astron. Soc. **526**, 5494 (2023), arXiv:2309.07959 [astro-ph.CO].
[28] M.-X. Lin, B. Jain, M. Raveri, E. J. Baxter, C. Chang, *et al.*, Late time modification of structure growth and the S_8 tension, Phys. Rev. D **109**, 063523 (2024), arXiv:2308.16183 [astro-ph.CO].
[29] E. V. Linder, Exploring the Expansion History of the Universe, Phys. Rev. Lett. **90**, 091301 (2003), arXiv:astro-ph/0208512 [astro-ph].
[30] M. Chevallier and D. Polarski, Accelerating Universes with Scaling Dark Matter, International Journal of Modern Physics D **10**, 213 (2001), arXiv:gr-qc/0009008 [gr-qc].
[31] R. Calderon, K. Lodha, A. Shafieloo, E. Linder, W. Sohn, *et al.*, DESI 2024: Reconstructing Dark Energy using Crossing Statistics with DESI DR1 BAO data, arXiv e-prints , arXiv:2405.04216 (2024), arXiv:2405.04216 [astro-ph.CO].
[32] J. Rebouças, D. H. F. de Souza, K. Zhong, V. Miranda, and R. Rosenfeld, Investigating late-time dark energy and massive neutrinos in light of DESI Y1 BAO, Journal of Cosmology and Astroparticle Physics **2025**, 024 (2025), arXiv:2408.14628 [astro-ph.CO].
[33] K. Lodha, R. Calderon, W. L. Matthewson, A. Shafieloo, M. Ishak, *et al.*, Extended dark energy analysis using DESI DR2 BAO measurements, Phys. Rev. D **112**, 083511 (2025), arXiv:2503.14743 [astro-ph.CO].
[34] M. Cortês and A. R. Liddle, On data set tensions and signatures of new cosmological physics, Mon. Not. Roy. Astron. Soc. **531**, L52 (2024), arXiv:2309.03286 [astro-ph.CO].
[35] R. Liu, Y. Zhu, W. Hu, and V. Miranda, Phantom Mirage from Axion Dark Energy, arXiv e-prints , arXiv:2510.14957 (2025), arXiv:2510.14957 [astro-ph.CO].
[36] S. Afroz and S. Mukherjee, Hint toward an inconsistency between BAO and supernovae datasets: The evidence of redshift evolving dark energy from DESI DR2 is absent, Phys. Rev. D **113**, 083514 (2026), arXiv:2504.16868 [astro-ph.CO].
[37] E. Silva, M. A. Sabogal, M. Scherer, R. C. Nunes, E. Di Valentino, and S. Kumar, New constraints on interacting dark energy from DESI DR2 BAO observations, Phys. Rev. D **111**, 123511 (2025), arXiv:2503.23225 [astro-ph.CO].
[38] A. Chakraborty, P. Chanda, S. Das, and K. Dutta, DESI results: hint towards coupled dark matter and dark energy, Journal of Cosmology and Astroparticle Physics **2025**, 047 (2025), arXiv:2503.10806 [astro-ph.CO].
[39] D. H. Lee, W. Yang, E. Di Valentino, S. Pan, and C. van de Bruck, Shape of dark energy: Constraining its evolution with a general parametrization, Phys. Rev. D **113**, 063554 (2026), arXiv:2507.11432 [astro-ph.CO].
[40] S. Pan, S. Paul, E. N. Saridakis, and W. Yang, Interacting dark energy after DESI DR2: A challenge for the ΛCDM paradigm?, Phys. Rev. D **113**, 023515 (2026), arXiv:2504.00994 [astro-ph.CO].
[41] A. Chudaykin and M. Kunz, Modified gravity interpretation of the evolving dark energy in light of DESI data, Phys. Rev. D **110**, 123524 (2024), arXiv:2407.02558 [astro-ph.CO].
[42] S. Goldstein, M. Celoria, and F. Schmidt, Monodromic dark energy and DESI, Phys. Rev. D **113**, 023532 (2026), arXiv:2507.16970 [astro-ph.CO].
[43] G. Ye, M. Martinelli, B. Hu, and A. Silvestri, Hints of Nonminimally Coupled Gravity in DESI 2024 Baryon Acoustic Oscillation Measurements, Phys. Rev. Lett. **134**, 181002 (2025), arXiv:2407.15832 [astro-ph.CO].
[44] Y. Tada and T. Terada, Quintessential interpretation of the evolving dark energy in light of DESI observations, Phys. Rev. D **109**, L121305 (2024), arXiv:2404.05722 [astro-ph.CO].
[45] M. Gökçen, Ö. Akarsu, and E. Di Valentino, Revisiting CPL with sign-switching density: To cross or not to cross the NECB, Physics of the Dark Universe **52**, 102273 (2026), arXiv:2602.21169 [astro-ph.CO].
[46] F. van Die and V. Desjacques, Sound Mode and Scale-Dependent Growth in Two-Fluid Dynamical Dark Energy, arXiv e-prints , arXiv:2603.05049 (2026), arXiv:2603.05049 [astro-ph.CO].
[47] B. Ratra and P. J. E. Peebles, Cosmological consequences of a rolling homogeneous scalar field, Phys. Rev. D **37**, 3406 (1988).
[48] R. R. Caldwell, R. Dave, and P. J. Steinhardt, Cosmological Imprint of an Energy Component with General Equation of State, Phys. Rev. Lett. **80**, 1582 (1998), arXiv:astro-ph/9708069 [astro-ph].
[49] T. Chiba, T. Okabe, and M. Yamaguchi, Kinetically driven quintessence, Phys. Rev. D **62**, 023511 (2000),

arXiv:astro-ph/9912463 [astro-ph].
[50] Y.-F. Cai, E. N. Saridakis, M. R. Setare, and J.-Q. Xia, Quintom cosmology: Theoretical implications and observations, Physics Reports **493**, 1 (2010), arXiv:0909.2776 [hep-th].
[51] A. Vikman, Can dark energy evolve to the phantom?, Phys. Rev. D **71**, 023515 (2005), arXiv:astro-ph/0407107 [astro-ph].
[52] P. Creminelli, G. D'Amico, J. Noreña, and F. Vernizzi, The effective theory of quintessence: the w ¡ -1 side unveiled, Journal of Cosmology and Astroparticle Physics **2009**, 018 (2009), arXiv:0811.0827 [astro-ph].
[53] S. Nojiri, S. D. Odintsov, and V. K. Oikonomou, Is Phantom Divide Crossing in General Relativity Completely Impossible? Shortcomings and Possible Solutions, arXiv e-prints , arXiv:2601.21356 (2026), arXiv:2601.21356 [gr-qc].
[54] P. Creminelli, G. D'Amico, J. Noreña, and F. Vernizzi, The effective theory of quintessence: the w ¡ -1 side unveiled, Journal of Cosmology and Astroparticle Physics **2009**, 018 (2009), arXiv:0811.0827 [astro-ph].
[55] G. Gubitosi, F. Piazza, and F. Vernizzi, The effective field theory of dark energy, Journal of Cosmology and Astroparticle Physics **2013**, 032 (2013), arXiv:1210.0201 [hep-th].
[56] N. Frusciante and L. Perenon, Effective field theory of dark energy: A review, Physics Reports **857**, 1 (2020), arXiv:1907.03150 [astro-ph.CO].
[57] G. Horndeski, Second-Order Scalar-Tensor Field Equations in a Four-Dimensional Space, International Journal of Theoretical Physics **10**, 363 (1974).
[58] E. Bellini and I. Sawicki, Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity, Journal of Cosmology and Astroparticle Physics **2014**, 050 (2014), arXiv:1404.3713 [astro-ph.CO].
[59] W. Hu, Structure Formation with Generalized Dark Matter, Astrophys. J. **506**, 485 (1998), arXiv:astro-ph/9801234 [astro-ph].
[60] H. Kodama and M. Sasaki, Cosmological Perturbation Theory, Prog. Theor. Phys. Suppl. **78**, 1 (1984).
[61] F. Hassani, J. Adamek, M. Kunz, and F. Vernizzi, k-evolution: a relativistic N-body code for clustering dark energy, Journal of Cosmology and Astroparticle Physics **2019**, 011 (2019), arXiv:1910.01104 [astro-ph.CO].
[62] L. Blot, T. Gayoux, F. Schmidt, P. S. Corasaniti, and B. de Ligondes, Cosmological hydrodynamical simulations of clustering dark energy with Nefertiti, arXiv e-prints , arXiv:2603.16248 (2026), arXiv:2603.16248 [astro-ph.CO].
[63] J. Dakin, S. Hannestad, T. Tram, M. Knabenhans, and J. Stadel, Dark energy perturbations in N-body simulations, Journal of Cosmology and Astroparticle Physics **2019**, 013 (2019), arXiv:1904.05210 [astro-ph.CO].
[64] G. Brando, K. Koyama, and D. Wands, Relativistic corrections to the growth of structure in modified gravity, Journal of Cosmology and Astroparticle Physics **2021**, 013 (2021), arXiv:2006.11019 [astro-ph.CO].
[65] G. Ballesteros and J. Lesgourgues, Dark energy with non-adiabatic sound speed: initial conditions and detectability, JCAP **10**, 014, arXiv:1004.5509 [astro-ph.CO].
[66] R. Bean and O. Doré, Probing dark energy perturbations: The dark energy equation of state and speed of sound as measured by WMAP, Phys. Rev. D **69**, 083503 (2004), arXiv:astro-ph/0307100 [astro-ph].
[67] E. Calabrese, R. de Putter, D. Huterer, E. V. Linder, and A. Melchiorri, Future CMB constraints on early, cold, or stressed dark energy, Phys. Rev. D **83**, 023011 (2011), arXiv:1010.5612 [astro-ph.CO].
[68] J. Xia, Y. Cai, T. Qiu, G.-B. Zhao, and X. Zhang, Constraints on the sound speed of dynamical dark energy, International Journal of Modern Physics D **17**, 1229 (2007).
[69] S. Hannestad, Constraints on the sound speed of dark energy, Physical Review D **71**, 103519 (2005).
[70] R. D. Putter, D. Huterer, and E. Linder, Measuring the speed of dark: Detecting dark energy perturbations, Physical Review D **81**, 103513 (2010).
[71] G. Brando, K. Koyama, D. Wands, M. Zumalacárregui, I. Sawicki, *et al.*, Fully relativistic predictions in Horndeski gravity from standard Newtonian N-body simulations, Journal of Cosmology and Astroparticle Physics **2021**, 024 (2021), arXiv:2105.04491 [astro-ph.CO].
[72] N. Shah, K. Koyama, and J. Noller, Dark energy constraints in light of theoretical priors, arXiv e-prints , arXiv:2507.19450 (2025), arXiv:2507.19450 [astro-ph.CO].
[73] C. Deffayet, O. Pujolas, I. Sawicki, and A. Vikman, Imperfect Dark Energy from Kinetic Gravity Braiding, JCAP **10**, 026, arXiv:1008.0048 [hep-th].
[74] M. Cataneo and K. Koyama, Non-parametric exploration of minimally coupled gravity with phantom crossing, arXiv e-prints , arXiv:2512.13691 (2025), arXiv:2512.13691 [astro-ph.CO].
[75] W. Fang, W. Hu, and A. Lewis, Crossing the phantom divide with parametrized post-Friedmann dark energy, Phys. Rev. D **78**, 087303 (2008), arXiv:0808.3125 [astro-ph].
[76] J. A. Frieman, C. T. Hill, A. Stebbins, and I. Waga, Cosmology with Ultralight Pseudo Nambu-Goldstone Bosons, Phys. Rev. Lett. **75**, 2077 (1995), arXiv:astro-ph/9505060 [astro-ph].
[77] A. de Felice and S. Tsujikawa, Cosmology of a Covariant Galileon Field, Phys. Rev. Lett. **105**, 111301 (2010), arXiv:1007.2700 [astro-ph.CO].
[78] A. Barreira, B. Li, C. M. Baugh, and S. Pascoli, The observational status of Galileon gravity after Planck, Journal of Cosmology and Astroparticle Physics **2014**, 059 (2014), arXiv:1406.0485 [astro-ph.CO].
[79] H. A. Buchdahl, Non-linear lagrangians and cosmological theory, Monthly Notices of the Royal Astronomical Society **150**, 1 (1970), https://academic.oup.com/mnras/article-pdf/150/1/1/8075909/mnras150-0001.pdf.
[80] W. Hu and I. Sawicki, Models of f(R) cosmic acceleration that evade solar system tests, Phys. Rev. D **76**, 064004 (2007), arXiv:0705.1158 [astro-ph].
[81] C. D. Kreisch and E. Komatsu, Cosmological constraints on Horndeski gravity in light of GW170817, Journal of Cosmology and Astroparticle Physics **2018**, 030 (2018), arXiv:1712.02710 [astro-ph.CO].
[82] D. Alonso, E. Bellini, P. G. Ferreira, and M. Zumalacárregui, Observational future of cosmological scalar-tensor theories, Phys. Rev. D **95**, 063502 (2017), arXiv:1610.09290 [astro-ph.CO].
[83] M. Cataneo and E. Bellini, mochi_class: Modelling Optimisation to Compute Horndeski In class, The Open

Journal of Astrophysics **7**, 76 (2024), arXiv:2407.11968 [astro-ph.CO].
[84] L. Amendola, M. Kunz, and D. Sapone, Measuring the dark side (with weak lensing), Journal of Cosmology and Astroparticle Physics **2008**, 013 (2008), arXiv:0704.2421 [astro-ph].
[85] R. Caldwell, A. Cooray, and A. Melchiorri, Constraints on a new post-general relativity cosmological parameter, Phys. Rev. D **76**, 023507 (2007), arXiv:astro-ph/0703375 [astro-ph].
[86] Planck Collaboration, P. A. R. Ade, N. Aghanim, M. Arnaud, M. Ashdown, *et al.*, Planck 2015 results. XIV. Dark energy and modified gravity, Astronomy & Astrophysics **594**, A14 (2016), arXiv:1502.01590 [astro-ph.CO].
[87] T. M. C. Abbott, M. Aguena, A. Alarcon, O. Alves, A. Amon, *et al.*, Dark Energy Survey Year 3 results: Constraints on extensions to Λ CDM with weak lensing and galaxy clustering, Phys. Rev. D **107**, 083504 (2023), arXiv:2207.05766 [astro-ph.CO].
[88] O. Sergijenko and B. Novosyadlyj, Sound speed of scalar field dark energy: Weak effects and large uncertainties, Physical Review D **91**, 083007 (2014).
[89] J. Carron, M. Mirmelstein, and A. Lewis, CMB lensing from Planck PR4 maps, Journal of Cosmology and Astroparticle Physics **2022**, 039 (2022), arXiv:2206.07773 [astro-ph.CO].
[90] J. Carron, A. Lewis, and G. Fabbian, Planck integrated Sachs-Wolfe-lensing likelihood and the CMB temperature, Phys. Rev. D **106**, 103507 (2022), arXiv:2209.07395 [astro-ph.CO].
[91] A. Amon, D. Gruen, M. A. Troxel, N. MacCrann, S. Dodelson, *et al.*, Dark Energy Survey Year 3 results: Cosmology from cosmic shear and robustness to data calibration, Phys. Rev. D **105**, 023514 (2022), arXiv:2105.13543 [astro-ph.CO].
[92] L. F. Secco, S. Samuroff, E. Krause, B. Jain, J. Blazek, *et al.*, Dark Energy Survey Year 3 results: Cosmology from cosmic shear and robustness to modeling uncertainty, Phys. Rev. D **105**, 023515 (2022), arXiv:2105.13544 [astro-ph.CO].
[93] M. Gatti, E. Sheldon, A. Amon, M. Becker, M. Troxel, *et al.*, Dark energy survey year 3 results: weak lensing shape catalogue, Monthly Notices of the Royal Astronomical Society **504**, 4312 (2021), arXiv:2011.03408 [astro-ph.CO].
[94] E. Krause and T. Eifler, cosmolike - cosmological likelihood analyses for photometric galaxy surveys, Monthly Notices of the Royal Astronomical Society **470**, 2100 (2017), arXiv:1601.05779 [astro-ph.CO].
[95] A. Stebbins, Weak Lensing On the Celestial Sphere, arXiv e-prints , astro-ph/9609149 (1996), arXiv:astro-ph/9609149 [astro-ph].
[96] C. M. Hirata and U. Seljak, Intrinsic alignment-lensing interference as a contaminant of cosmic shear, Phys. Rev. D **70**, 063526 (2004), arXiv:astro-ph/0406275 [astro-ph].
[97] M. L. Brown, A. N. Taylor, N. C. Hambly, and S. Dye, Measurement of intrinsic alignments in galaxy ellipticities, Monthly Notices of the Royal Astronomical Society **333**, 501 (2002), arXiv:astro-ph/0009499 [astro-ph].
[98] A. J. Mead, S. Brieden, T. Tröster, and C. Heymans, HMCODE-2020: improved modelling of non-linear cosmological power spectra with baryonic feedback, Monthly Notices of the Royal Astronomical Society **502**, 1401 (2021), arXiv:2009.01858 [astro-ph.CO].
[99] C. Trendafilova, A. R. Khalife, and S. Galli, The end of easy phenomenology for CMB experiments: A case study in the dark sector, Journal of Cosmology and Astroparticle Physics **2025**, 094 (2025), arXiv:2502.19383 [astro-ph.CO].
[100] A. Albrecht, G. Bernstein, R. Cahn, W. L. Freedman, J. Hewitt, *et al.*, Report of the Dark Energy Task Force, arXiv e-prints , astro-ph/0609591 (2006), arXiv:astro-ph/0609591 [astro-ph].
[101] N. Shah, K. Koyama, and J. Noller, Dark energy constraints in light of theoretical priors, JCAP **01**, 054, arXiv:2507.19450 [astro-ph.CO].
[102] S. Tassev, M. Zaldarriaga, and D. Eisenstein, Solving Large Scale Structure in Ten Easy Steps with COLA, JCAP **06**, 036, arXiv:1301.0322 [astro-ph.CO].
[103] J. Gordon, B. F. de Aguiar, J. Rebouças, G. Brando, F. Falciano, *et al.*, Modeling nonlinear scales with the comoving Lagrangian acceleration method: Preparing for LSST Y1, Phys. Rev. D **110**, 083529 (2024), arXiv:2404.12344 [astro-ph.CO].
[104] J. Rebouças, V. Lloyd, J. Gordon, G. Brando, and V. Miranda, Modeling nonlinear scales for dynamical dark energy cosmologies with COLA, arXiv e-prints , arXiv:2510.14888 (2025), arXiv:2510.14888 [astro-ph.CO].